

the series. Since

$$\int_0^z \frac{d\zeta}{(\zeta - k)^2} = -\left(\frac{1}{z - k} + \frac{1}{k}\right), \quad k \neq 0,$$

$$\int_0^z \left(\frac{\pi^2}{\sin^2(\pi\zeta)} - \frac{1}{\zeta^2}\right) d\zeta = -\pi \cot(\pi z) + \frac{1}{z},$$

we obtain from (2.1) the partial fractions decomposition

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{k \neq 0} \left(\frac{1}{z - k} + \frac{1}{k}\right).$$

If we combine the terms for $\pm k$, the constants cancel, and we obtain

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{k=1}^{\infty} \left(\frac{1}{z - k} + \frac{1}{z + k}\right) = \frac{1}{z} + 2z \sum_{k=1}^{\infty} \frac{1}{z^2 - k^2}.$$

Exercises for XIII.2

1. Use the partial fractions decomposition of $\pi^2/\sin^2(\pi z)$ to establish the formula

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

2. Establish the partial fractions decomposition

$$\pi \tan(\pi z) = -2z \sum_{n=0}^{\infty} \frac{1}{z^2 - \left(n + \frac{1}{2}\right)^2}.$$

Hint. Use $\tan w = \cot w - 2 \cot(2w)$.

3. Show that

$$\frac{\pi}{\sin(\pi z)} = \frac{1}{z} + 2z \sum_{n=1}^{\infty} \frac{(-1)^n}{z^2 - n^2}.$$

4. Show that

$$\frac{\pi}{\cos(\pi z)} = \sum_{n=1}^{\infty} \frac{(-1)^n (2n - 1)}{z^2 - \left(n - \frac{1}{2}\right)^2}.$$

5. Let $\{z_k\}$ be a sequence of distinct points such that $|z_k| \rightarrow \infty$ and $\sum |z_k|^{-m-1} < \infty$. Show that $z^m \sum 1/z_k^m (z - z_k)$ converges normally to a meromorphic function with principal part $1/(z - z_k)$ at z_k . (If $z_k = 0$, we replace the corresponding summand by $1/z$.)
6. Construct a meromorphic function on the complex plane whose poles are simple poles at the Gaussian integers $m + ni$ with residue 1.

7. Construct a meromorphic function on the complex plane whose poles are double poles at the points $\log n$, $n \geq 1$, with principal parts $1/(z - \log n)^2 + 1/(z - \log n)$.
8. Construct a meromorphic function on the open unit disk $\mathbb{D}$ whose poles are simple poles at the points $(1 - 2^{-n})e^{2\pi ik/n}$, $1 \leq k \leq n$, $n \geq 1$, with residue 1.
9. Show that $\sum_{-\infty}^{\infty} 1/(z^3 - n^3)$ converges normally to a meromorphic function. Locate the poles and find the corresponding principal parts of the function. Express the function in terms of trigonometric functions (specifically, the cotangent function).
10. Show that the lattice points $m\omega_1 + n\omega_2$, $-\infty < m, n < \infty$, can be arranged in a sequence $\{z_k\}_{k=0}^{\infty}$ such that $|z_k| \geq c\sqrt{k}$.
11. Let $\{z_k\}$ be a sequence of distinct points such that $|z_k| > c\sqrt{k}$. Show that

$$\sum \left[\frac{1}{(z - z_k)^2} - \frac{1}{z_k^2} \right]$$

converges normally on $\mathbb{C}$ and absolutely at each $z \in \mathbb{C}$.

12. Let $f(z)$ be a doubly periodic meromorphic function on $\mathbb{C}$ with periods ω_1 and ω_2 , and let $\mathcal{P}(z)$ be the Weierstrass P -function associated with the periods ω_1 and ω_2 . (a) Show that if the only poles of $f(z)$ are double poles at the lattice points $m\omega_1 + n\omega_2$, $-\infty < m, n < \infty$, then there are constants a and b such that $f(z) = a\mathcal{P}(z) + b$. (b) Show that if the only poles of $f(z)$ are triple poles at the lattice points $m\omega_1 + n\omega_2$, $-\infty < m, n < \infty$, then there are constants a, b, c such that $f(z) = a\mathcal{P}(z) + b\mathcal{P}'(z) + c$. (c) Show that $\mathcal{P}'(z)^2 = 4\mathcal{P}(z)^3 + a\mathcal{P}(z)^2 + b\mathcal{P}(z) + c$ for some constants a, b, c .
13. Let D be a domain in $\mathbb{C}$, and let $E_k = \{|z - z_k| \leq r_k\}$, $k \geq 1$, be disjoint closed disks in D that accumulate only on the boundary of D . Suppose $Q_k(z)$ is analytic for $|z - z_k| > r_k$. Show that there is an analytic function $f(z)$ on $D \setminus \bigcup_{k=1}^{\infty} E_k$ such that for each k , $f(z) - Q_k(z)$ extends analytically to E_k .

3. Infinite Products

An **infinite product** is an expression of the form $\prod_{j=1}^{\infty} p_j$, where the p_j 's are complex numbers. We say that the infinite product **converges** if $p_j \rightarrow 1$ and $\sum \text{Log } p_j$ converges, where we sum only over terms for which $p_j \neq 0$. If the infinite product converges, we define its value to be 0 if one

of the p_j 's is 0; otherwise, we define it to be

$$\prod_{j=1}^{\infty} p_j = \exp \left(\sum_{j=1}^{\infty} \text{Log } p_j \right).$$

Thus any question we might ask about infinite products can be translated to a question about infinite series by taking logarithms.

To help clarify the definition of convergent infinite product, we make several simple observations. First, if $\prod p_j$ converges, then at most finitely many of the p_j 's can be 0. This is because $p_j \rightarrow 1$. Second, if $\prod p_j$ converges, then

$$\prod_{j=1}^{\infty} p_j = \lim_{m \rightarrow \infty} \prod_{j=1}^m p_j = \lim_{m \rightarrow \infty} p_1 p_2 \cdots p_m.$$

This is because $\exp(\text{Log } p_j) = p_j$. Third, we can always factor out a finite number of terms from a convergent infinite product,

$$\prod_{j=1}^{\infty} p_j = p_1 p_2 \cdots p_N \prod_{j=N+1}^{\infty} p_j.$$

Finally, if an infinite product converges, and none of the factors are 0, then the product cannot be 0.

Example. Consider

$$\prod_{k=1}^{\infty} \left(1 + \frac{(-1)^{k+1}}{k} \right) = (1+1) \left(1 - \frac{1}{2} \right) \left(1 + \frac{1}{3} \right) \left(1 - \frac{1}{4} \right) \cdots$$

Since $(1 + 1/(2k-1))(1 - 1/(2k)) = 1$, the product of the first m terms is equal to 1 if m is even, and it is equal to the last factor $1 + 1/m$ if m is odd. Thus the product converges to 1.

It is often convenient to write $p_j = 1 + a_j$ and to express the product as $\prod(1 + a_j)$. If the product converges, then $a_j \rightarrow 0$, and only finitely many of the a_j 's are equal to -1 . For most purposes we can ignore the terms for which $1 + a_j = 0$ and work with the “tail” $\prod_{j=N}^{\infty} (1 + a_j)$ of the infinite product, for which a_j is near 0.

If $0 < t \leq 1$, we have the estimate $t/2 \leq \log(1+t) \leq t$. From this estimate it follows that if $t_j \geq 0$, then $\sum t_j$ converges if and only if $\sum \text{Log}(1 + t_j)$ converges. This leads immediately to the following test for convergence of infinite products.

Theorem. If $t_j \geq 0$, then $\prod(1+t_j)$ converges if and only if $\sum t_j$ converges.

Example. The infinite product

$$\prod_{k=1}^{\infty} \left(1 + \frac{1}{k^{\alpha}}\right) = (1+1) \left(1 + \frac{1}{2^{\alpha}}\right) \left(1 + \frac{1}{3^{\alpha}}\right) \dots$$

converges for $\alpha > 1$ and diverges for $\alpha \leq 1$.

The infinite product $\prod(1+a_j)$ is said to **converge absolutely** if $a_j \rightarrow 0$ and $\sum \text{Log}(1+a_j)$ converges absolutely, where we sum over the terms for which $a_j \neq -1$. If $\prod(1+a_j)$ converges absolutely, then $\sum \text{Log}(1+a_j)$ converges, and $\prod(1+a_j)$ converges.

Since $\text{Log}(1+w)$ is analytic at $w=0$ and has power series expansion $\text{Log}(1+w) = w + \mathcal{O}(w^2)$, we see that $|\text{Log}(1+a_j)|$ is comparable to $|a_j|$ when a_j is near 0. Consequently, $\sum |\text{Log}(1+a_j)|$ converges if and only if $\sum |a_j|$ converges. This together with the preceding theorem yield the following.

Theorem. *The infinite product $\prod(1+a_j)$ converges absolutely if and only if $\sum a_j$ converges absolutely. This occurs if and only if $\prod(1+|a_j|)$ converges.*

Example. We have seen that $\prod(1+(-1)^{k+1}/k)$ converges. However, it does not converge absolutely, on account of the divergence of the harmonic series, $\sum 1/k = \infty$.

Example. Consider the infinite product $\prod(1+i/k)$. Since $\text{Log}(1+i/k) = i/k + \mathcal{O}(1/k^2)$, the series $\sum \text{Log}(1+i/k)$ does not converge, by comparison with the harmonic series. Consequently, $\prod(1+i/k)$ does not converge. However, since $0 < \text{Log}|1+i/k| = \frac{1}{2} \log(1+1/k^2) < 1/k^2$, the infinite product $\prod |1+i/k|$ *does* converge. Thus absolute convergence of an infinite product is not equivalent to the convergence of the product of the absolute values of the factors.

Now we turn to infinite products of functions. The Weierstrass M -test for a sum of functions is converted easily to the following test for an infinite product of functions.

Theorem. *Suppose that $g_k(x) = 1 + h_k(x)$, $k \geq 1$, are functions on a set E . Suppose that there are constants $M_k > 0$ such that $\sum M_k < \infty$, and $|h_k(x)| \leq M_k$ for $x \in E$. Then $\prod_{k=1}^m g_k(x)$ converges to $\prod_{k=1}^{\infty} g_k(x)$ uniformly on E as $m \rightarrow \infty$.*

Choose a constant C such that $|\log(1+w)| \leq C|w|$ for $|w| \leq \frac{1}{2}$, and choose N such that $M_k \leq \frac{1}{2}$ for $k \geq N$. The condition $|h_k(x)| \leq M_k \leq \frac{1}{2}$ implies that $|\text{Log}(1+h_k(x))| \leq CM_k$. By the Weierstrass M -test, the series $\sum_{k=N}^{\infty} \text{Log}(1+h_k(x))$ converges uniformly on E . If we exponentiate we obtain uniform convergence of the partial products $\prod_{k=N}^m g_k(x)$ to

$\prod_{k=N}^{\infty} g_k(x)$ as $m \rightarrow \infty$. Since each $g_k(x)$ is bounded, when we multiply by the first $N - 1$ factors we obtain uniform convergence of $\prod_{k=1}^m g_k(x)$ to $\prod_{k=1}^{\infty} g_k(x)$ as $m \rightarrow \infty$.

If $G(z) = g_1(z) \cdots g_m(z)$ is a finite product of analytic functions, then by taking logarithms and differentiating, we obtain

$$(3.1) \quad \frac{G'(z)}{G(z)} = \frac{g_1'(z)}{g_1(z)} + \cdots + \frac{g_m'(z)}{g_m(z)}.$$

This procedure is called **logarithmic differentiation**. The logarithmic differentiation formula also holds for uniformly convergent infinite products of analytic functions. It is proved by applying the formula above to finite subproducts and passing to the limit.

Theorem. Let $g_k(z)$, $k \geq 1$, be analytic functions on a domain D such that $\prod_{k=1}^m g_k(z)$ converges normally on D to $G(z) = \prod_{k=1}^{\infty} g_k(z)$. Then

$$(3.2) \quad \frac{G'(z)}{G(z)} = \sum_{k=1}^{\infty} \frac{g_k'(z)}{g_k(z)}, \quad z \in D,$$

where the sum converges normally on D .

Note that the function $G'(z)/G(z)$ has poles at the zeros of $G(z)$. However, the hypothesis implies that $g_k(z) \rightarrow 1$ uniformly on any compact subset of D , so the summands $g_k'(z)/g_k(z)$ are analytic on the compact subset for k large. Since the uniform convergence of a series is not affected by the first terms of the series, the poles do not affect the uniform convergence. Since normal convergence of a sequence of analytic functions implies uniform convergence of the derivatives of the functions in the sequence, we may apply (3.1) to the partial product $G_m(z) = g_1(z) \cdots g_m(z)$ and pass to the limit, to obtain (3.2).

Example. Consider the infinite product $f(z) = z \prod_{k=1}^{\infty} (1 - z^2/k^2)$. By the Weierstrass M -test, the series $\sum |z|^2/k^2$ converges uniformly on any bounded set. Thus the infinite product converges uniformly on any bounded set, and $f(z)$ is an entire function. Since the zeros of $f(z)$ are simple zeros at the integers, we suspect that $f(z)$ is related to $\sin(\pi z)$. To check this out, we differentiate logarithmically, to obtain

$$\frac{f'(z)}{f(z)} = \frac{1}{z} + 2z \sum_{k=1}^{\infty} \frac{1}{z^2 - k^2} = \frac{1}{z} + 2z \sum_{k=1}^{\infty} \left[\frac{1}{z - k} + \frac{1}{z + k} \right].$$

This expression we recognize from Section 2 as the partial fractions decomposition of $\pi \cot(\pi z) = \pi \cos(\pi z)/\sin(\pi z)$. We integrate, and we see that $\log f(z)$ is $\log \sin(\pi z)$ up to adding a constant. Hence $f(z) = C \sin(\pi z)$. Since $f(z)/z \rightarrow 1$ as $z \rightarrow 0$, we have $C = 1/\pi$. Thus we obtain an infinite

product expansion for the sine function,

$$(3.3) \quad \sin(\pi z) = \pi z \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{k^2}\right) = \pi z (1 - z^2) \left(1 - \frac{z^2}{4}\right) \cdots$$

Exercises for XIII.3

1. Evaluate the following.

$$(a) \prod_{n=1}^{\infty} \left(1 + \frac{1}{n(n+2)}\right) \quad (b) \prod_{n=2}^{\infty} \left(1 - \frac{1}{n^2}\right) \quad (c) \prod_{n=3}^{\infty} \left(\frac{n^2-1}{n^2-4}\right)$$

2. Define $a_k = -\frac{1}{\sqrt{k}}$ if k is odd, and $a_k = \frac{1}{\sqrt{k}} + \frac{1}{k} + \frac{1}{k\sqrt{k}}$ if k is even. Show that $\prod(1 + a_k)$ converges, while $\sum a_k$ and $\sum a_k^2$ diverge.

3. Show that if $t_j \geq 0$, then $\prod(1 + t_j) \leq \exp(\sum t_j)$.

4. Show that if $0 < t_j < 1$, then $\prod(1 - t_j)$ converges if and only if $\sum t_j$ converges.

5. Show that the infinite product $\prod(1 + a_j)$ converges if and only if there is $N \geq 1$ such that $\lim_{m \rightarrow \infty} \prod_{j=N}^m (1 + a_j)$ exists and is nonzero.

6. Show that $\prod(1 + a_j)$ converges if and only if $\prod_{j=m}^n (1 + a_j) \rightarrow 1$ as $m, n \rightarrow \infty$. *Hint.* Take logarithms and invoke the Cauchy criterion for series.

7. Show that if $\prod(1 + a_k)$ converges, then $\prod |1 + a_k|$ converges.

8. Suppose $a_k \rightarrow 0$. Show that the series $\sum a_k$ converges absolutely if and only if both the series $\sum \text{Arg}(1 + a_k)$ and $\sum \text{Log} |1 + a_k|$ converge absolutely.

9. Show that $\prod_{n=1}^{\infty} \left(1 + \frac{1}{n^2}\right) = \frac{e^{\pi} - e^{-\pi}}{2\pi}$.

10. Show that $\prod_{n=0}^{\infty} (1 + z^{2^n}) = \frac{1}{1-z}$ for $|z| < 1$.

11. Show that if $p_k(z)$ is a polynomial of degree k such that $p_k(0) = 1$ and $p_k(z)$ has no zeros in the disk $\{|z| \leq k^3\}$, then $\prod p_k(z)$ converges normally.

12. Establish one of the following formulae, and deduce from it the other using logarithmic differentiation:

$$e^z - 1 = ze^{z/2} \prod_{k=1}^{\infty} \left(1 + \frac{z^2}{4\pi^2 k^2} \right),$$

$$\frac{1}{e^z - 1} = \frac{1}{z} - \frac{1}{2} + 2z \sum_{k=1}^{\infty} \frac{1}{z^2 + 4\pi^2 k^2}.$$

13. Use the infinite product expansion for $\sin(\pi z)$ to show that the **Wallis product**

$$\prod_{k=1}^{\infty} \frac{(2k)^2}{(2k-1)(2k+1)} = \lim_{n \rightarrow \infty} \frac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdots (2n) \cdot (2n)}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots (2n-1) \cdot (2n+1)}$$

converges to $\pi/2$. Use this to show that

$$\lim_{n \rightarrow \infty} \frac{[n!]^2}{(2n)!} \frac{2^{2n}}{\sqrt{n}} = \sqrt{\pi}.$$

14. Show that if $t > 0$, then $\prod_{-m \leq k \leq tm} \left(1 + \frac{z}{k} \right)$ converges to $\frac{\sin(\pi z)}{\pi z} t^z$ as $m \rightarrow \infty$.

15. Show that $\frac{1}{z} \prod_{n=1}^{\infty} \frac{n}{z+n} \left(\frac{n+1}{n} \right)^z$ converges to a meromorphic function $\Gamma(z)$ whose poles are simple poles at 0 and the negative integers. Show that

$$\Gamma(z) = \lim_{m \rightarrow \infty} \frac{(m-1)! m^z}{z(z+1) \cdots (z+m-1)}.$$

Show that $\Gamma(z+1) = z\Gamma(z)$. Show that $\Gamma(n+1) = n!$ for positive integers n . *Remark.* The function $\Gamma(z)$ is called the **gamma function**. It was first introduced by Euler, who defined it to be the limit above. We will give an equivalent definition in the next chapter.

16. Let α_k be a sequence of complex numbers, with possible repetitions, such that $|\alpha_k| < 1$ and $|\alpha_k| \rightarrow 1$, and consider the **infinite Blaschke product** defined by

$$B(z) = \prod \frac{\overline{\alpha_k}}{|\alpha_k|} \frac{\alpha_k - z}{1 - \overline{\alpha_k} z},$$

where the factors corresponding to $\alpha_k = 0$ are z .

- (a) Suppose that $\sum(1 - |\alpha_k|) < \infty$. Let E be the set of accumulation points on the unit circle $\partial\mathbb{D}$ of the α_k 's. Show that the infinite product converges uniformly on compact subsets

of $\mathbb{C}^* \setminus E$ to a meromorphic function $B(z)$, with the following properties: $|B(z)| < 1$ for $z \in \mathbb{D}$, $|B(z)| = 1$ for $z \in \partial\mathbb{D} \setminus E$, and $B(z)$ has zeros precisely at the points α_k .

- (b) Show that if $\sum(1 - |\alpha_k|) = +\infty$, then the partial products converge uniformly on compact subsets of $\mathbb{D}$ to 0.
- (c) Suppose that $f(z)$ is a bounded analytic function on $\mathbb{D}$ that is not identically zero. Show that $f(z)$ has a factorization $f(z) = B(z)g(z)$, where $B(z)$ is a (finite or infinite) Blaschke product, and $g(z)$ is a bounded analytic function on $\mathbb{D}$ with no zeros. In particular, the zeros $\alpha_1, \alpha_2, \dots$ of $f(z)$, repeated according to multiplicity, satisfy $\sum(1 - |\alpha_k|) < +\infty$.

4. The Weierstrass Product Theorem

The Weierstrass product theorem is a companion theorem to the Mittag-Leffler theorem. The Mittag-Leffler theorem asserts that we can prescribe the poles and principal parts of a meromorphic function. The Weierstrass product theorem asserts that we can prescribe the zeros and poles of a meromorphic function together with their orders.

Recall that the order of a meromorphic function $f(z)$ at a point z_0 is the order of the zero if $f(z_0) = 0$, and it is minus the order of the pole if $f(z)$ has a pole at z_0 . If z_0 is neither a pole nor a zero of $f(z)$, the order of $f(z)$ at z_0 is defined to be 0.

Theorem (Weierstrass Product Theorem). *Let D be a domain in the complex plane. Let $\{z_k\}$ be a sequence of distinct points of D with no accumulation point in D , and let $\{n_k\}$ be a sequence of integers (positive or negative). Then there is a meromorphic function $f(z)$ on D whose only zeros and poles are at the points z_k , such that the order of $f(z)$ at z_k is n_k .*

The proof runs parallel to the proof of the Mittag-Leffler theorem. Let K_m be the set of $z \in D$ such that $|z| \leq m$ and the distance from z to ∂D is at least $1/m$. Then K_m is a compact subset of D , $K_m \subset K_{m+1}$, and each component of $\mathbb{C}^* \setminus K_m$ contains a point of $\mathbb{C}^* \setminus D$. Suppose $z_k \in K_{m+1} \setminus K_m$. We connect z_k to a point $w_k \in \mathbb{C}^* \setminus D$ by a simple curve γ_k in $\mathbb{C}^* \setminus K_m$. If $w_k \neq \infty$ we define $f_k(z)$ to be an analytic branch of $\log((z - z_k)/(z - w_k))$ in the simply connected domain $\mathbb{C}^* \setminus \gamma_k$. If $w_k = \infty$, we take $f_k(z)$ to be an analytic branch of $\log(1 - z/z_k)$. By Runge's theorem, there is a rational function $g_k(z)$ with only pole at w_k such that $|f_k(z) - g_k(z)| \leq 2^{-k}/n_k$ on K_m . We consider the product

$$f(z) = \prod_{k=1}^{\infty} \left(\frac{z - z_k}{z - w_k} \right)^{n_k} e^{-n_k g_k(z)},$$

where we replace the factor in parentheses by $1 - z/z_k$ if $w_k = \infty$ and $z_k \neq 0$ and by z^{n_k} if both $w_k = \infty$ and $z_k = 0$. Now $n_k(f_k(z) - g_k(z))$ is not defined on γ_k . However, its exponential

$$\exp[n_k(f_k(z) - g_k(z))] = \left(\frac{z - z_k}{z - w_k} \right)^{n_k} e^{-n_k g_k(z)}$$

is meromorphic on D and has order n_k at z_k and no other poles or zeros. By the Weierstrass M -test, $\sum_{k=N}^{\infty} n_k(f_k(z) - g_k(z))$ converges uniformly on K_m , where N is chosen sufficiently large to exclude terms with $z_k \in K_m$. Hence the infinite product defining $f(z)$ converges normally on D to a meromorphic function. Clearly, $f(z)$ has the desired zeros and poles.

In the case that $D = \mathbb{C}$ is the entire complex plane, the points w_k are all ∞ . Suppose $z_k \neq 0$ for $k \geq 1$, and let n_0 be the order (possibly zero) of $f(z)$ at $z = 0$. The product expansion then has the form

$$f(z) = z^{n_0} \prod_{k=1}^{\infty} \left(1 - \frac{z}{z_k} \right)^{n_k} e^{-n_k g_k(z)},$$

where $g_k(z)$ is an approximation to $\text{Log}(1 - z/z_k)$ for $|z| < |z_k|$. We can take $g_k(z)$ to be a partial sum for the power series expansion of $\text{Log}(1 - z/z_k)$ at $z = 0$. Since

$$\text{Log}(1 - \zeta) = -\zeta - \frac{\zeta^2}{2} - \frac{\zeta^3}{3} - \cdots - \frac{\zeta^m}{m} + \mathcal{O}(\zeta^{m+1}),$$

we can take

$$g_k(z) = - \left(\frac{z}{z_k} + \frac{z^2}{2z_k^2} + \frac{z^3}{3z_k^3} + \cdots + \frac{z^{m_k}}{m_k z_k^{m_k}} \right),$$

where m_k is chosen large enough to guarantee convergence of the product.

Exercise. Find an entire function with simple zeros at the negative integers and no other zeros.

Solution. In this case the product

$$\prod_{k=1}^{\infty} \left(1 + \frac{z}{k} \right)$$

does not converge. However, from the estimate

$$\left| \text{Log} \left(1 + \frac{z}{k} \right) - \frac{z}{k} \right| \leq C \frac{|z|^2}{k^2}, \quad |z| \leq R, \quad k \geq 2R,$$

we see that

$$(4.1) \quad \prod_{k=1}^{\infty} \left(1 + \frac{z}{k} \right) e^{-z/k}$$

converges normally on the entire complex plane to an entire function with the prescribed zeros.

Exercises for XIII.4

1. Let $G(z)$ be the entire function defined by the infinite product (4.1). Show that $\pi z G(z) G(-z) = \sin(\pi z)$.
2. Construct an entire function that has simple zeros at the points n^2 , $n \geq 0$, and no other zeros.
3. Construct an entire function that has simple zeros on the real axis at the points $\pm n^{1/4}$, $n \geq 0$, and no other zeros.
4. Construct an entire function that has simple zeros on the positive real axis at the points $\sqrt{n}$, $n \geq 1$, and double zeros on the imaginary axis at the points $\pm i\sqrt{n}$, $n \geq 1$, and no other zeros.
5. Construct an entire function that has simple zeros at the Gaussian integers $m + ni$, $-\infty < m, n < \infty$, and no other zeros.
6. Find *all* entire functions $f(z)$ that satisfy the functional equation $f(2z) = (1 - 2z)f(z)$. Express the answer in terms of an infinite product.
7. Show that

$$\left| \operatorname{Log}(1 - \zeta) + \zeta + \frac{\zeta^2}{2} + \frac{\zeta^3}{3} + \cdots + \frac{\zeta^N}{N} \right| \leq \frac{1}{N} \frac{|\zeta|^{N+1}}{1 - |\zeta|},$$

for $|\zeta| < 1$.

8. Let $\{z_k\}$ be a sequence of distinct nonzero points such that $|z_k| \rightarrow \infty$. Let $N \geq 0$, and let $\{m_k\}$ be a sequence of positive integers such that $\sum m_k |z_k|^{-N-1} < \infty$. Show that

$$\prod_{k=1}^{\infty} \left(1 - \frac{z}{z_k}\right)^{m_k} \exp \left\{ m_k \left[\frac{z}{z_k} + \frac{z^2}{2z_k^2} + \cdots + \frac{z^N}{Nz_k^N} \right] \right\}$$

converges normally to an entire function with zeros of order m_k at z_k and no other zeros. *Hint.* See the estimate in the preceding exercise.

9. Show that any meromorphic function $f(z)$ on a domain D is the quotient $f(z) = g(z)/h(z)$ of two analytic functions $g(z)$ and $h(z)$ on D .
10. Let $f(z)$ be a meromorphic function on a simply connected domain D . Show that the meromorphic functions with the same zeros and poles of the same orders as $f(z)$ are precisely the functions of the form $f(z)e^{h(z)}$, where $h(z)$ is analytic on D .
11. Give a brief solution of Exercise 1.5 on interpolating sequences based on the Mittag-Leffler theorem and the Weierstrass product theorem.

XIV

Some Special Functions

Our aim in this chapter is to illustrate the power of complex analysis by proving a deep theorem in number theory, the prime number theorem, which does not appear at first glance to be related to complex analysis. Along the way we introduce various functions that play an important role in complex analysis. In Section 1 we introduce the gamma function $\Gamma(z)$, which provides a meromorphic extension of the factorial function. We derive the asymptotic properties of the gamma function in Section 2 by viewing it as a Laplace transform. This yields Stirling's asymptotic formula for $n!$. In Section 3 we study the zeta function, which is a meromorphic function whose zeros are related to the asymptotic distribution of prime numbers. In Section 4 we study Dirichlet series associated with various number-theoretic functions, thereby giving a strong hint of the fecund relationship between complex analysis and number theory. The proof of the prime number theorem is given in Section 5.

1. The Gamma Function

The gamma function $\Gamma(z)$ is a meromorphic function that arises frequently in complex analysis. It extends the factorial function from the positive integers to the entire complex plane. It is defined in the right half-plane by

$$(1.1) \quad \Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt, \quad \operatorname{Re} z > 0.$$

The integral defining $\Gamma(z)$ is absolutely convergent, and

$$|\Gamma(x + iy)| \leq \Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt, \quad x > 0.$$

Since the integrand depends analytically on the parameter z , the function $\Gamma(z)$ is analytic on the right half-plane.

To derive a functional equation for $\Gamma(z)$, we integrate by parts,

$$\Gamma(z+1) = \int_0^\infty e^{-t} t^z dt = - \int_0^\infty t^z d(e^{-t}) = - [t^z e^{-t}]_0^\infty + z \int_0^\infty e^{-t} t^{z-1} dt.$$

The term $t^z e^{-t}$ vanishes at 0 and at ∞ (technically, we integrate from 0 to R and take a limit as $R \rightarrow \infty$), and the identity becomes

$$(1.2) \quad \Gamma(z+1) = z\Gamma(z), \quad \operatorname{Re} z > 0.$$

To evaluate $\Gamma(z)$ at the positive integers, we begin with $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$, and we use the functional equation to obtain successively $\Gamma(2) = 1$, $\Gamma(3) = 2 \cdot 1$, $\Gamma(4) = 3 \cdot 2 \cdot 1$, and by induction

$$(1.3) \quad \Gamma(n+1) = n!, \quad n \geq 1.$$

We declare that $0! = \Gamma(1) = 1$, and then (1.3) also holds for $n = 0$.

The functional equation (1.2) allows us to extend $\Gamma(z)$ to the left half-plane, as follows. We apply the functional equation m times to obtain $\Gamma(z+m) = (z+m-1) \cdots (z+1)z\Gamma(z)$, which we rewrite as

$$\Gamma(z) = \frac{\Gamma(z+m)}{(z+m-1) \cdots (z+1)z}.$$

The right-hand side is defined and meromorphic for $\operatorname{Re} z > -m$, with simple poles at $z = 0, -1, \dots, -m+1$. By the uniqueness principle, the meromorphic extension is unique and it satisfies the functional equation (1.2). Passing to the limit as $m \rightarrow +\infty$, we obtain the following.

Theorem. *The function $\Gamma(z)$ extends to be meromorphic on the entire complex plane, where it satisfies the functional equation $\Gamma(z+1) = z\Gamma(z)$. Its poles are simple poles at $z = 0, -1, -2, \dots$.*

Now we will express $\Gamma(z)$ in terms of an infinite product. Define

$$\Gamma_n(z) = \int_0^n t^{z-1} \left(1 - \frac{t}{n}\right)^n dt, \quad \operatorname{Re} z > 0, \quad n \geq 1.$$

Since $(1 - t/n)^n \leq e^{-t}$ and $(1 - t/n)^n \rightarrow e^{-t}$ for $t \geq 0$, we have

$$\lim_{n \rightarrow \infty} \Gamma_n(z) = \Gamma(z).$$

(The easiest way to justify passing to the limit is to appeal to Lebesgue's dominated convergence theorem.) The substitution $s = t/n$ yields

$$(1.4) \quad \Gamma_n(z) = n^z \int_0^1 s^{z-1} (1-s)^n ds, \quad \operatorname{Re} z > 0, \quad n \geq 1.$$

Thus

$$\Gamma_1(z) = \int_0^1 s^{z-1} (1-s) ds = \left[\frac{s^z}{z} - \frac{s^{z+1}}{z+1} \right]_0^1 = \frac{1}{z} - \frac{1}{z+1} = \frac{1}{z(z+1)}.$$

If we integrate by parts in (1.4), we obtain

$$\begin{aligned}\Gamma_n(z) &= \frac{n^z}{z} \int_0^1 (1-s)^n d(s^z) = -\frac{n^z}{z} \int_0^1 s^z d(1-s)^n \\ &= \frac{n^{z+1}}{z} \int_0^1 s^z (1-s)^{n-1} ds = \left(\frac{n}{n-1}\right)^z \frac{n}{z} \Gamma_{n-1}(z+1).\end{aligned}$$

We repeat this step $n-1$ times and use the formula for $\Gamma_1(z)$, and we obtain

$$\Gamma_n(z) = \frac{n^z n!}{z(z+1) \cdots (z+n-2)} \Gamma_1(z+n-1) = \frac{n^z n!}{z(z+1) \cdots (z+n)}.$$

We rewrite this as

$$\frac{1}{\Gamma_n(z)} = \frac{1}{n^z} z(1+z) \left(1 + \frac{z}{2}\right) \cdots \left(1 + \frac{z}{n}\right).$$

We compare this with the infinite product with zeros at the negative integers,

$$G(z) = \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right) e^{-z/k},$$

which was discussed in Section XIII.4. The exponential convergence factors guarantee that the product converges normally on the entire complex plane. In order to express $1/\Gamma(z)$ in terms of $G(z)$, we write

$$\frac{1}{\Gamma_n(z)} = ze^{\gamma_n z} \prod_{k=1}^n \left(1 + \frac{z}{k}\right) e^{-z/k},$$

where

$$\gamma_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n.$$

We have seen (Section V.1) that the sequence $\{\gamma_n\}$ decreases to Euler's constant $\gamma = 0.5772 \dots$ as $n \rightarrow \infty$. Thus when we pass to the limit we obtain the following.

Theorem. *Let γ denote Euler's constant. Then*

$$(1.5) \quad \frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right) e^{-z/k}, \quad z \in \mathbb{C}.$$

Initially, the identity is established only in the right half-plane. However, both sides are meromorphic, so the identity persists for all $z \in \mathbb{C}$, by the uniqueness principle.

If we take logarithms in (1.5) and differentiate, we obtain

$$-\frac{\Gamma'(z)}{\Gamma(z)} = \frac{1}{z} + \gamma - \sum_{k=1}^{\infty} \frac{z/k^2}{1+z/k} = \frac{1}{z} + \gamma - z \sum_{k=1}^{\infty} \frac{1}{k} \frac{1}{z+k}.$$

Another differentiation leads to the expression

$$(1.6) \quad \frac{d}{dz} \frac{\Gamma'(z)}{\Gamma(z)} = \sum_{k=0}^{\infty} \frac{1}{(z+k)^2},$$

which we will return to in the next section.

There is another formula we require for later use, which follows from the infinite product expansion derived for $\sin(\pi z)$ in Section XIII.3.3. In terms of the function $G(z)$, it can be expressed (Exercise XIII.3.1) as $\sin(\pi z) = \pi z G(z) G(-z)$. Combining this with (1.5) we obtain

$$\frac{1}{\Gamma(z)\Gamma(-z)} = -z^2 G(z) G(-z) = -\frac{z \sin(\pi z)}{\pi}.$$

Since $\Gamma(1-z) = -z\Gamma(-z)$, this yields

$$(1.7) \quad \Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}, \quad z \in \mathbb{C}.$$

Exercises for XIV.1

1. Show that $\Gamma(z)$ has no zeros.
2. Show that for $m \geq 0$, the residue of $\Gamma(z)$ at $z = -m$ is $(-1)^m/m!$.
3. Show that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Use this to show that

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \sqrt{\pi} = 2^{1-2n} \frac{\Gamma(2n)}{\Gamma(n)} \sqrt{\pi}.$$

4. Use the Wallis product (Exercise XIII.3.13) to show that

$$\lim_{n \rightarrow \infty} 2^{2n-1} \sqrt{n} \frac{\Gamma(n)^2}{\Gamma(2n)} = \sqrt{\pi}.$$

5. Prove **Legendre's duplication formula**,

$$\Gamma(2z) = \frac{2^{2z-1}}{\sqrt{\pi}} \Gamma(z) \Gamma\left(z + \frac{1}{2}\right).$$

Hint. Show first that $\Gamma(2z)/\Gamma(z)\Gamma(z + \frac{1}{2})$ has the form Ae^{Bz} by separating the product for $\Gamma(2z)$ into even and odd terms; then evaluate A and B .

6. Show that

$$\lim_{n \rightarrow \infty} \frac{\Gamma(n + \frac{1}{2})}{\sqrt{n} \Gamma(n)} = 1.$$

7. Show that

$$\int_0^1 t^{p-1}(1-t)^{q-1}dt = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}, \quad p, q > 0.$$

Remark. This integral is called the **beta function**, denoted by $B(p, q)$. To establish the identity, begin by expressing $\Gamma(p)\Gamma(q)$ as a double integral and converting to polar coordinates. The r -integral yields $\Gamma(p+q)$ and the θ -integral leads to the integral above, with $t = \cos^2 \theta$.

2. Laplace Transforms

Our aim in this section is to derive an asymptotic approximation to Laplace transforms of functions that do not grow too fast. This leads to an asymptotic approximation to $\Gamma(z)$, thence to Stirling's formula giving an asymptotic approximation to $n!$.

Let $h(s)$, $s \geq 0$, be a continuous or piecewise continuous function on the positive real axis. The **Laplace transform of $h(s)$** is the function

$$(2.1) \quad (\mathcal{L}h)(z) = \int_0^\infty e^{-sz} h(s) ds.$$

provided that the integral converges. We will always assume that $h(s)$ has at most exponential growth, that is, there are constants B, C such that

$$|h(s)| \leq Ce^{Bs}, \quad 0 \leq s < \infty.$$

Then the integral in (2.1) converges absolutely and defines an analytic function in the half-plane $\{\operatorname{Re} z > B\}$. The estimate

$$|(\mathcal{L}h)(x + iy)| \leq C \int_0^\infty e^{-xs} e^{Bs} ds = \frac{C}{x - B}, \quad x > B,$$

shows that $(\mathcal{L}h)(z)$ is bounded on the half-plane $\{\operatorname{Re} z > B + \varepsilon\}$ for any $\varepsilon > 0$, and $(\mathcal{L}h)(z) \rightarrow 0$ as $\operatorname{Re} z \rightarrow \infty$.

Example. Suppose $h(s) = s^k$ for some integer $k \geq 0$. Since $s^k e^{-\varepsilon s} \rightarrow 0$ as $s \rightarrow \infty$ (polynomials grow more slowly than exponentials), there is for any $\varepsilon > 0$ an estimate of the form $|h(s)| \leq C_\varepsilon e^{\varepsilon s}$. Hence the Laplace transform $(\mathcal{L}s^k)(z)$ converges absolutely for all z satisfying $\operatorname{Re} z > 0$. We compute that

$$(\mathcal{L}s^k)(x) = \int_0^\infty e^{-xs} s^k ds = \frac{1}{x^{k+1}} \int_0^\infty e^{-t} t^k dt, \quad x > 0.$$

The integral on the right is $\Gamma(k+1) = k!$, so $(\mathcal{L}s^k)(z)$ coincides with $k!/z^{k+1}$ when $z = x > 0$. By the uniqueness principle it coincides with $k!/z^{k+1}$ on

the right half-plane,

$$(2.2) \quad (\mathcal{L}s^k)(z) = \frac{k!}{z^{k+1}}, \quad \operatorname{Re} z > 0, \quad k \geq 0.$$

Lemma. Suppose that $|h(s)| \leq Cs^N e^{Bs}$ for some integer $N \geq 0$ and constants B, C . Then

$$(2.3) \quad |(\mathcal{L}h)(x+iy)| = \mathcal{O}\left(\frac{1}{x^{N+1}}\right), \quad \text{as } x \rightarrow \infty.$$

This follows from the obvious estimates

$$|(\mathcal{L}h)(x+iy)| \leq C \int_0^\infty e^{-(x-B)s} s^N ds = C \frac{N!}{(x-B)^{N+1}} = \mathcal{O}\left(\frac{1}{x^{N+1}}\right).$$

Theorem. Suppose that $|h(s)| \leq Ce^{Bs}$ for $s \geq 0$. Fix $N \geq 1$. If $h(s)$ has the Taylor polynomial approximation

$$(2.4) \quad h(s) = b_0 + b_1 s + \frac{b_2}{2!} s^2 + \cdots + \frac{b_{N-1}}{(N-1)!} s^{N-1} + \mathcal{O}(s^N)$$

as $s \rightarrow 0+$, then

$$(2.5) \quad (\mathcal{L}h)(z) = \frac{b_0}{z} + \frac{b_1}{z^2} + \frac{b_2}{z^3} + \cdots + \frac{b_{N-1}}{z^N} + \mathcal{O}\left(\frac{1}{z^{N+1}}\right)$$

as $x = \operatorname{Re} z \rightarrow \infty$.

To see this, we let $R(s)$ be the remainder term in the Taylor approximation, so that

$$h(s) = b_0 + b_1 s + \cdots + \frac{b_{N-1}}{(N-1)!} s^{N-1} + R(s), \quad s \geq 0.$$

Since $|R(s)| \leq C_0 s^N$ for $s > 0$ near 0, there is $C_1 > 0$ such that

$$|R(s)| \leq C_1 s^N e^{(B+1)s}, \quad 0 \leq s < \infty.$$

Since the Laplace transform operates linearly,

$$(\mathcal{L}h)(z) = b_0(\mathcal{L}1)(z) + b_1(\mathcal{L}s)(z) + \cdots + \frac{b_{N-1}}{(N-1)!} (\mathcal{L}s^{N-1})(z) + (\mathcal{L}R)(z).$$

The result now follows from (2.2) and (2.3).

We wish to use the asymptotic approximation theorem to obtain asymptotic approximations to $\Gamma(z)$. We begin by expressing the derivative of $\Gamma'(z)/\Gamma(z)$ as a Laplace transform. From $(\mathcal{L}s)(z) = 1/z^2$ we obtain

$$\frac{1}{(z+k)^2} = (\mathcal{L}s)(z+k) = \int_0^\infty e^{-s(z+k)} s ds = \int_0^\infty e^{-sz} (e^{-s})^k s ds.$$

Summing over k from 0 to N , and recognizing the geometric series, we obtain

$$\sum_{k=1}^N \frac{1}{(z+k)^2} = \int_0^\infty e^{-sz} \sum_{k=0}^N (e^{-s})^k s ds = \int_0^\infty e^{-sz} \frac{1 - e^{-s(N+1)}}{1 - e^{-s}} s ds.$$

The exponentials $e^{-s(N+1)}$ decrease to 0, and passage to the limit in the right-hand side is justified. In view of (1.6), we obtain in the limit as $N \rightarrow \infty$ that

$$\frac{d}{dz} \frac{\Gamma'(z)}{\Gamma(z)} = \int_0^\infty e^{-sz} \frac{s}{1 - e^{-s}} ds, \quad \operatorname{Re} z > 0.$$

The integral on the right is the Laplace transform of the function

$$g(s) = \frac{s}{1 - e^{-s}}.$$

This function $g(s)$ is a meromorphic function of the complex variable s , with poles at $s = 2\pi ni$ for $n = \pm 1, \pm 2, \dots$. The function is analytic at $s = 0$, since $1 - e^{-s}$ has a simple zero there. The power series expansion of $g(s)$ about $s = 0$ is readily calculated to be

$$g(s) = 1 + \frac{1}{2}s + \frac{1}{12}s^2 + \frac{g^{(4)}(0)}{4!}s^4 + \dots, \quad |s| < 2\pi.$$

Since $g(s)/s \rightarrow 1$ as $s \rightarrow +\infty$, an estimate of the form $0 \leq g(s) \leq Ce^s$, $s \geq 0$, holds for some constant $C > 0$. The theorem then yields the asymptotic approximation

$$\frac{d}{dz} \frac{\Gamma'(z)}{\Gamma(z)} = \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{6z^3} + \mathcal{O}\left(\frac{1}{x^5}\right), \quad x = \operatorname{Re} z \rightarrow \infty.$$

To obtain an asymptotic approximation to $\Gamma(z)$, we must integrate twice. Except for a constant of integration, the antiderivative for any error term $\mathcal{O}(1/x^N)$, $N \geq 2$, is dominated by

$$\int_{x+iy}^\infty \frac{dt}{|t+iy|^N} \leq \int_x^\infty \frac{dt}{t^N} = \frac{1}{(N-1)x^{N-1}} = \mathcal{O}\left(\frac{1}{x^{N-1}}\right), \quad N \geq 2.$$

Hence

$$\frac{\Gamma'(z)}{\Gamma(z)} = \operatorname{Log} z + \alpha - \frac{1}{2z} - \frac{1}{12z^2} + \mathcal{O}\left(\frac{1}{x^4}\right),$$

where α is a constant of integration. If we integrate again, we obtain

$$\log \Gamma(z) = z \operatorname{Log} z - z + \alpha z - \frac{1}{2} \operatorname{Log} z + \beta + \frac{1}{12z} + \mathcal{O}\left(\frac{1}{x^3}\right),$$

where β is another constant of integration. If we substitute the asymptotic expansions for $\log \Gamma(z+1)$ and $\log \Gamma(z)$ into the relation $\log \Gamma(z+1) = \log z + \log \Gamma(z)$, we obtain $\alpha = 0$. To evaluate β , we compare

$\Gamma(2n)$ with $\Gamma(n)$, which can be done with the elementary relation in Exercise 1.3(c).

$$\Gamma(n)\Gamma\left(n + \frac{1}{2}\right) = 2^{2n-1}\Gamma(2n)\sqrt{\pi}.$$

If we take logarithms, substitute the approximations, and do some book-keeping, we obtain $\beta = \log \sqrt{2\pi}$. We have proved the following.

Theorem. *The gamma function $\Gamma(z)$ satisfies*

$$(2.6) \quad \log \Gamma(z) = \left(z - \frac{1}{2}\right) \text{Log } z - z + \log \sqrt{2\pi} + \frac{1}{12z} + \mathcal{O}\left(\frac{1}{x^3}\right)$$

as $x = \text{Re } z \rightarrow \infty$.

The asymptotic approximation to $\Gamma(z)$ is obtained by exponentiating.

$$\Gamma(z) = z^z e^{-z} \sqrt{\frac{2\pi}{z}} \left(1 + \frac{1}{12z} + \mathcal{O}\left(\frac{1}{x^2}\right)\right), \quad x = \text{Re } z \rightarrow \infty.$$

Substituting $n! = n\Gamma(n)$, we obtain **Stirling's formula**,

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \left(1 + \frac{1}{12n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right).$$

It is natural to consider the asymptotic behavior of Laplace transforms in the right half-plane as $x = \text{Re } z \rightarrow \infty$. However, for some classes of functions it is more appropriate to consider asymptotic behavior in a sector as $|z| \rightarrow \infty$ rather than in a half-plane. Let $f(z)$ be defined and analytic in a sector $S = \{-\theta_0 < \arg z < \theta_1\}$. We say that $\sum_{j=0}^{\infty} a_j/z^j$ is an **asymptotic series** for $f(z)$ in the sector S if for each $N \geq 0$,

$$f(z) - \sum_{j=0}^N \frac{a_j}{z^j} = \mathcal{O}\left(\frac{1}{z^{N+1}}\right)$$

as $z \rightarrow \infty$ in the sector. If $\sum a_j/z^j$ is an asymptotic series for $f(z)$, then

$$a_k = \lim_{z \rightarrow \infty} z^k \left[f(z) - a_0 - \frac{a_1}{z} - \cdots - \frac{a_{k-1}}{z^{k-1}} \right].$$

Hence the coefficients in the asymptotic series of $f(z)$ are uniquely determined. We refer to $\sum a_j/z^j$ as the **asymptotic expansion** of $f(z)$ in the sector.

If $f(z)$ is analytic at ∞ , then its Laurent series is an asymptotic series for $f(z)$ in any sector. However, asymptotic series generally do not converge. (See the Exercises.) Further, it can happen that different functions have the same asymptotic expansion. For example, since $z^n e^{-z} \rightarrow 0$ as $z \rightarrow \infty$ in any sector $\{|\arg z| < (\pi/2) - \varepsilon\}$, the function e^{-z} has an asymptotic series in the sector that is identically zero.

The asymptotic approximation theorem shows that if $h(s)$ has at most exponential growth, and if $h(s)$ is infinitely differentiable at $s = 0$, then the Laplace transform $(\mathcal{L}h)(z)$ has the asymptotic expansion $\sum h^{(k)}(0)/z^{k+1}$ in any sector $S_\varepsilon = \{|\arg z| < (\pi/2) - \varepsilon\}$. In any such sector there is an estimate of the form $c_\varepsilon|z| \leq \operatorname{Re} z \leq |z|$; consequently, $|z| \rightarrow \infty$ if and only if $x = \operatorname{Re} z \rightarrow \infty$. In general, the series need not be an asymptotic expansion for $(\mathcal{L}h)(z)$ in the entire right half-plane, regarded as a sector. However, in the case of the function $\log \Gamma(z)$, it is. In fact, it can be shown that the series (2.6) is an asymptotic series for $\operatorname{Log} \Gamma(z)$ in any sector $\{|\arg z| < \pi - \varepsilon\}$, for any $\varepsilon > 0$.

Exercises for XIV.2

1. Show that if $\operatorname{Re} \lambda > -1$, then the Laplace transform of s^λ converges absolutely in the right half-plane, and

$$(\mathcal{L}s^\lambda)(z) = \frac{\Gamma(\lambda+1)}{z^{\lambda+1}}, \quad \operatorname{Re} z > 0.$$

2. Show that if $g(s) = e^{as}h(s)$, then $(\mathcal{L}g)(z) = (\mathcal{L}h)(z-a)$.
3. Suppose $h(s) = 0$ for $s < 0$. Show that if $a > 0$ and $g(s) = h(s-a)$, then $(\mathcal{L}g)(z) = e^{-az}(\mathcal{L}h)(z)$.
4. Show that if $h(s)$ is differentiable, then the Laplace transform of its derivative $h'(s)$ is $(\mathcal{L}h')(z) = z(\mathcal{L}h)(z) - h(0)$.
5. Compute $\left(\frac{n}{e}\right)^n \sqrt{2\pi n}$ and $\left(\frac{n}{e}\right)^n \sqrt{2\pi n} \left(1 + \frac{1}{12n}\right)$ for $n = 2, 3$, and 4, and compare the results with $n!$.
6. Define the **Bernoulli numbers** $B_1, B_2, B_3, \dots$ by

$$1 - \frac{z}{2} \cot \frac{z}{2} = \frac{B_1}{2!} z^2 + \frac{B_2}{4!} z^4 + \frac{B_3}{6!} z^6 + \dots$$

- (a) Show that $B_1 = 1/6$ and $B_2 = 1/30$. (b) Show that

$$\frac{s}{1 - e^{-s}} = 1 + \frac{1}{2}s + \frac{B_1}{2!}s^2 - \frac{B_2}{4!}s^4 + \frac{B_3}{6!}s^6 - \dots$$

- (c) Show that $\log \Gamma(z)$ has asymptotic series

$$\log \Gamma(z) \sim (z - 1/2) \log z - z + \log \sqrt{2\pi} + \frac{B_1}{2} \frac{1}{z} - \frac{B_2}{3 \cdot 4} \frac{1}{z^3} + \frac{B_3}{5 \cdot 6} \frac{1}{z^5} - \dots$$

- (d) Show that as $z \rightarrow \infty$ through the sector $|\arg z| \leq \pi/2 - \varepsilon$,

$$\Gamma(z) = z^z e^{-z} \sqrt{\frac{2\pi}{z}} \left(1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51840z^3} + \mathcal{O}\left(\frac{1}{|z|^4}\right) \right).$$

How many terms can be specified here using only B_1 and B_2 ?

7. Express the function

$$F(x) = \int_x^\infty \frac{e^{x-t}}{t} dt, \quad x > 0,$$

as a Laplace transform, and show that it extends analytically to the right half-plane. Obtain the asymptotic expansion of the function, and show that the asymptotic expansion diverges at every point.

8. Let $h(s)$, $s \geq 0$, be a continuous function such that $|h(s)| \leq Ce^{Bs}$ for some constants $B, C > 0$. Suppose that $h(s)$ extends to be analytic in a neighborhood of the origin, and suppose that the power series expansion $h(s) = \sum_{k=0}^\infty h^{(k)}(0)s^k/k!$ has only a finite radius of convergence. Show that the asymptotic series $\sum_{k=0}^\infty h^{(k)}(0)/z^{k+1}$ of the Laplace transform $(\mathcal{L}h)(z)$ diverges at every point z .
9. Show that if $f(z)$ is analytic at ∞ , then its power series expansion in descending powers of z is an asymptotic series for $f(z)$ in any sector.
10. Show that if $f(z)$ and $g(z)$ have asymptotic expansions in a sector, then so does their product $f(z)g(z)$.
11. Show that if $f(z)$ has an asymptotic series $\sum_{k=2}^\infty a_k/z^k$ in a sector S , beginning with the $1/z^2$ term, then any antiderivative $F(z)$ of $f(z)$ has an asymptotic series in S obtained by integrating term by term.
12. Show that if $f(z)$ has an asymptotic series $\sum_{k=0}^\infty a_k/z^k$ in a sector $S = \{\theta_0 < \arg z < \theta_1\}$, then $f'(z)$ has the asymptotic series $\sum (-ja_j)/z^{j+1}$, obtained by differentiating term by term, in any smaller sector $S_\varepsilon = \{\theta_0 + \varepsilon < \arg z < \theta_1 - \varepsilon\}$.

3. The Zeta Function

The zeta function was first introduced by Euler. He derived an infinite product representation for the zeta function that connected it to prime numbers. A century later Riemann discovered a close relationship between the zeta function and the asymptotic distribution of prime numbers. Except for certain “trivial” zeros on the negative axis, the zeros of the zeta function are easily seen to lie in the “critical” strip $\{0 \leq \operatorname{Re} s \leq 1\}$. Riemann related the zeros of the zeta function to the asymptotic distribution of prime numbers by an “explicit formula,” and he conjectured that the zeros in the critical strip all lie on the line $\{\operatorname{Re} s = \frac{1}{2}\}$. At the beginning of the new millennium the most famous unsolved problem in complex analysis, if not in all of mathematics, is to determine whether the Riemann hypothesis holds. We will show in the next section that the zeta function

has no zeros on the boundary lines of the critical strip, and this will lead to the prime number theorem.

In this context it is traditional to denote the complex variable by $s = \sigma + it$. With this notation, the **zeta function** is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \sigma = \operatorname{Re} s > 1.$$

The series converges absolutely for $\sigma > 1$, and it converges uniformly for $\sigma \geq 1 + \varepsilon$ for any $\varepsilon > 0$, by comparison with the numerical series $\sum n^{-(1+\varepsilon)}$. The function $\zeta(\sigma)$ is a decreasing function of σ for $1 < \sigma < \infty$, and $\zeta(\sigma) \rightarrow 1$ as $\sigma \rightarrow +\infty$. Since the harmonic series $\sum 1/n$ diverges, $\zeta(\sigma) \rightarrow +\infty$ as $\sigma \rightarrow 1+$. Note also that $|\zeta(\sigma + it)| \leq \zeta(\sigma)$ for all t . We have already met the series for $\zeta(2)$,

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

For the remainder of this chapter, we will always use p to denote a prime number, $p = 2, 3, 5, 7, 11, \dots$. The following infinite product representation of the zeta function signals a connection between $\zeta(s)$ and prime numbers.

Theorem. *If $\operatorname{Re} s > 1$, then*

$$(3.1) \quad \frac{1}{\zeta(s)} = \prod_p \left(1 - \frac{1}{p^s}\right),$$

where the product is taken over all prime numbers p .

The series $\sum 1/p^{\sigma+it}$ converges absolutely for $\sigma > 1$, and it converges uniformly in any half-plane $\{\sigma \geq 1 + \varepsilon\}$. Hence the product (3.1) converges. Consider the geometric series

$$\frac{1}{1 - p^{-s}} = 1 + p^{-s} + p^{-2s} + \dots, \quad \operatorname{Re} s > 1.$$

If we multiply together the m series corresponding to primes $p_1, \dots, p_m$, we obtain

$$\frac{1}{(1 - p_1^{-s}) \cdots (1 - p_m^{-s})} = \sum_{k_1, \dots, k_m=0}^{\infty} (p_1^{k_1} \cdots p_m^{k_m})^{-s}.$$

Since every integer $n \geq 1$ has a unique representation as a product of powers of distinct primes, a summand $1/n^s$ appears at most once in this sum, so the sum is a subsum of the series $\sum n^{-s}$. As we incorporate more primes into the product, we eventually capture all terms n^{-s} , and in the limit we have

$$\prod_p \frac{1}{1 - p^{-s}} = \sum_{n=1}^{\infty} n^{-s} = \zeta(s).$$

Next we wish to extend $\zeta(s)$ to be a meromorphic function on the entire complex plane. To do this, we represent $\zeta(s)$ as a contour integral. We make a branch cut along the positive real axis and consider the function

$$(-z)^{s-1} = e^{(s-1)\operatorname{Log}(-z)}, \quad z \in \mathbb{C} \setminus [0, \infty),$$

which depends analytically on both z and the complex parameter s . Let γ be the contour indicated above, following the top edge of the branch cut from $+\infty$ to ε , detouring along a circle of radius ε around $z = 0$, and returning from ε to $+\infty$ along the bottom edge of the branch cut. We consider the function

$$(3.2) \quad \phi(s) = \frac{1}{2\pi i} \int_{\gamma} \frac{(-z)^{s-1}}{e^z - 1} dz.$$

The integral converges, and it represents an entire function of s . To evaluate it, we first assume that $\operatorname{Re} s > 1$, and we express

$$\begin{aligned} \phi(s) &= \frac{1}{2\pi i} \int_{\infty}^{\varepsilon} \frac{e^{(s-1)(\log x - i\pi)}}{e^x - 1} dx + \frac{1}{2\pi i} \int_{|z|=\varepsilon} \frac{(-z)^{s-1}}{e^z - 1} dz \\ &\quad + \frac{1}{2\pi i} \int_{\varepsilon}^{\infty} \frac{e^{(s-1)(\log x + i\pi)}}{e^x - 1} dx. \end{aligned}$$

Since $e^z - 1$ has a simple zero at $z = 0$, the integrand is bounded on the circle $|z| = \varepsilon$ by $C\varepsilon^{\operatorname{Re} s - 2}$, and the second integral is bounded by $C\varepsilon^{\operatorname{Re} s - 1}$. Under our assumption that $\operatorname{Re} s > 1$, this integral tends to 0 with ε . Passing to the limit, we obtain

$$\phi(s) = \frac{1}{2\pi i} \left(e^{i\pi(s-1)} - e^{-i\pi(s-1)} \right) \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx.$$

The term in parentheses is $2i \sin(\pi(s-1)) = -2i \sin(\pi s)$. Hence

$$(3.3) \quad \phi(s) = -\frac{\sin(\pi s)}{\pi} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx.$$

We treat this integral by expanding $1/(e^x - 1) = e^{-x}/(1 - e^{-x})$ in a geometric series,

$$(3.4) \quad \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx = \int_0^{\infty} \left(\sum_{n=1}^{\infty} e^{-nx} \right) x^{s-1} dx.$$

The partial sums of the geometric series form an increasing sequence of functions that converges uniformly on each interval $[\varepsilon, \infty)$. From this it is straightforward to justify interchanging the summation and integration. Since

$$\int_0^{\infty} e^{-nx} x^{s-1} dx = \frac{1}{n^s} \int_0^{\infty} e^{-t} t^{s-1} dt = \frac{\Gamma(s)}{n^s},$$

(3.4) becomes

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1} dx = \Gamma(s)\zeta(s), \quad \operatorname{Re} s > 1.$$

Substituting this into (3.3) and using $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ from (1.7), we obtain

$$\phi(s) = -\frac{\sin(\pi s)}{\pi}\Gamma(s)\zeta(s) = -\frac{\zeta(s)}{\Gamma(1-s)}.$$

We have established this identity for $\operatorname{Re} s > 1$. However, $\phi(s)$ is an entire function, so we can use this identity to extend $\zeta(s)$ to be a meromorphic function in the entire complex plane. The extended function then satisfies the following.

Theorem. *For the branch of $(-z)^{s-1}$ and the contour γ described above, we have*

$$(3.5) \quad \zeta(s) = -\Gamma(1-s)\phi(s) = -\frac{\Gamma(1-s)}{2\pi i} \int_\gamma \frac{(-z)^{s-1}}{e^z - 1} dz.$$

Since $\phi(s)$ is an entire function, and the poles of $\Gamma(1-s)$ are simple poles at $s = 1, 2, 3, \dots$, $\zeta(s)$ is analytic except for possibly a simple pole at $s = 1$. Now, $\Gamma(s)$ has a simple pole at $s = 0$ with residue 1, so $\Gamma(1-s) \sim 1/(1-s)$ as $s \rightarrow 1$. Since

$$\phi(1) = \frac{1}{2\pi i} \int_\gamma \frac{dz}{e^z - 1} = \operatorname{Res} \left[\frac{1}{e^z - 1}, 1 \right] = 1,$$

we see that $\zeta(s) \sim 1/(s-1)$ as $s \rightarrow 1$. Thus $\zeta(s)$ has a simple pole with residue 1 at $s = 1$. We collect these observations.

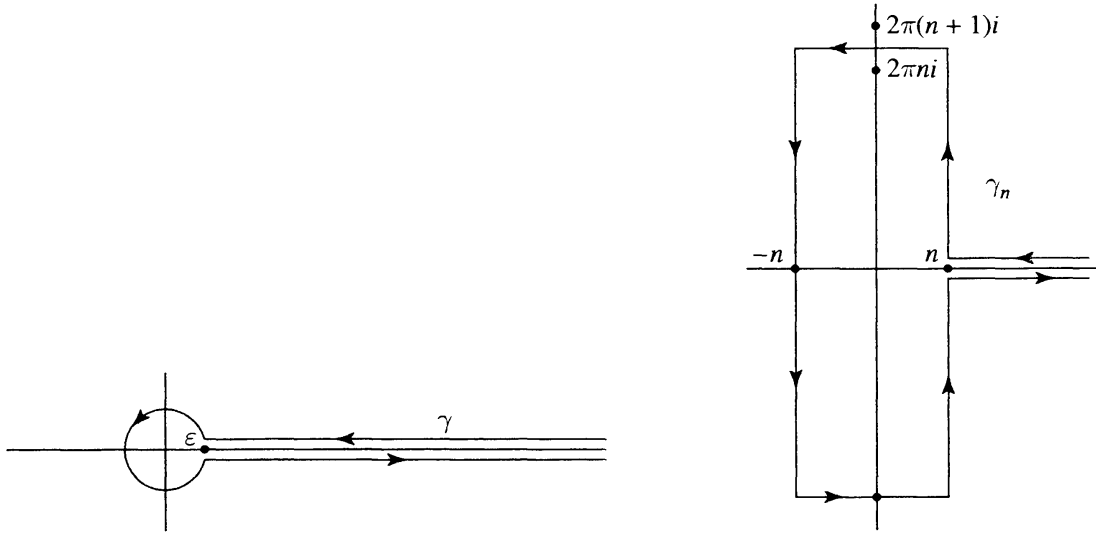
Theorem. *The zeta function $\zeta(s)$ is a meromorphic function on the complex plane with only one pole, a simple pole at $s = 1$ with residue 1.*

Next we derive a functional equation for the zeta function.

Theorem. *The function $\zeta(s)$ satisfies the functional equation*

$$(3.6) \quad \zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

For the proof, we modify the contour γ used to define $\phi(s)$. Fix s to be real and negative. Let γ_n be the contour indicated in the figure above, obtained from γ by replacing the segment starting at n on the top edge of the slit and going around 0 and back along the bottom edge of the slit to n



by straight line segments running counterclockwise around the boundary of the rectangle R_n with vertices at $\pm n \pm (2n + 1/2)\pi i$. Define

$$\phi_n(s) = \frac{1}{2\pi i} \int_{\gamma_n} \frac{(-z)^{s-1}}{e^z - 1} dz.$$

Since $|e^z - 1| > \frac{1}{2}$ on the edges of the rectangle, the integrand is bounded on the edges by $2n^{s-1}$. By the ML -estimate, the integrals over these edges are bounded by Cn^s . Since $s < 0$, this tends to 0 as $n \rightarrow \infty$, and consequently, $\phi_n(s) \rightarrow 0$ as $n \rightarrow \infty$. Now, the difference $\int_{\gamma_n} - \int_{\gamma}$ is the integral around the closed contour consisting of the boundary of the rectangle R_n with a detour along an elongated indentation that does not cross over the branch cut. The contour encloses only the poles of the integrand at the points $\pm 2\pi ki$, $0 < k \leq n$. Each pole is a simple pole, with residue

$$\left. \frac{(-z)^{s-1}}{e^z} \right|_{z=\pm 2\pi ki} = (2\pi)^{s-1} |k|^{s-1} e^{(s-1)\text{Log}(\pm i)}.$$

If we combine the residues for $\pm k$ and sum, we obtain from the residue theorem

$$\phi_n(s) - \phi(s) = (2\pi)^{s-1} \left(e^{(s-1)i\pi/2} + e^{-(s-1)i\pi/2} \right) \sum_{k=1}^n k^{s-1}.$$

We substitute

$$e^{(s-1)i\pi/2} + e^{-(s-1)i\pi/2} = 2 \cos\left((s-1)\frac{\pi}{2}\right) = 2 \sin\left(\frac{\pi s}{2}\right)$$

and let $n \rightarrow \infty$. Since $\phi_n(s) \rightarrow 0$, we obtain in the limit

$$-\phi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \zeta(1-s).$$

If we combine this with (3.5), we obtain the functional equation (3.6), at least for $s < 0$. Since both sides of the identity (3.6) are meromorphic, (3.6) holds for all s , by the uniqueness principle.

The functional equation of the zeta function yields information on its zeros. On account of the product representation (3.1), the zeta function has no zeros in the half-plane $\{\operatorname{Re} s > 1\}$. Since $1/\Gamma(z)$ has a product representation expressing it as an entire function, $\Gamma(z)$ has no zeros. The functional equation then shows that the only zeros of $\zeta(s)$ in the left half-plane $\{\operatorname{Re} s < 0\}$ are the zeros of $\sin(\pi s/2)$, which are simple zeros at $s = -2, -4, -6, \dots$. As mentioned above, these zeros are called the **trivial zeros** of the zeta function. The nontrivial zeros of the zeta function lie in the strip $\{0 \leq \operatorname{Re} s \leq 1\}$, which is called the **critical strip**. Many outstanding mathematicians have worked on the **Riemann hypothesis**, that the nontrivial zeros of the zeta function lie on the line $\{\operatorname{Re} s = \frac{1}{2}\}$. It is known that $\zeta(s)$ has infinitely many zeros in the critical strip. Their asymptotic distribution is known, and it is known that at least a third of them lie on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$. Further, the first zillion or so zeros are known to lie on the critical line. Meanwhile, whether the Riemann hypothesis is true remains a famous unresolved problem.

Exercises for XIV.3

1. Show that if a is a zero of the zeta function in the critical strip, then so are $\bar{a}$, $1 - a$, and $1 - \bar{a}$.
2. Show that the function $\xi(s) = \frac{1}{2}s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is an entire function that is symmetric with respect to the line $\{\operatorname{Re} s = \frac{1}{2}\}$, that is, $\xi(1-s) = \xi(s)$. Show that $\xi(s)$ is real when s is real and also when $\operatorname{Re} s = \frac{1}{2}$. Where are the zeros of $\xi(s)$?
3. By comparing $\zeta(\sigma)$ with $\int_1^\infty x^{-\sigma} dx$, show directly from the definition of $\zeta(s)$ that $(\sigma - 1)\zeta(\sigma) \rightarrow 1$ as $\sigma \rightarrow 1+$.
4. Show that $(1 - 2^{1-s})\zeta(s)$ is an entire function, which is represented by the series

$$(1 - 2^{1-s})\zeta(s) = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots, \quad \operatorname{Re} s > 1.$$

Remark. We will see in the next section that the series converges and represents the function for $\operatorname{Re} s > 0$.

5. By comparing the partial fractions decomposition

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z - n} + \frac{1}{n} \right)$$

with the series defining the Bernoulli numbers, and differentiating, derive the power series expansion

$$\sum_{n \neq 0} \frac{1}{(z - n)^2} = \sum_{k=0}^{\infty} \frac{2^{2k+1} \pi^{2k+2} B_{k+1}}{(2k)!(k+1)} z^{2k}, \quad |z| < 1.$$

Use this to show that

$$\zeta(2m) = \frac{2^{2m-1}\pi^{2m}B_m}{(2m)!}, \quad m \geq 1.$$

4. Dirichlet Series

A **Dirichlet series** is a series of the form

$$(4.1) \quad \sum_{n=1}^{\infty} \frac{a_n}{n^s} = a_1 + \frac{a_2}{2^s} + \frac{a_3}{3^s} + \frac{a_4}{4^s} + \cdots,$$

which we will often abbreviate to $\sum a_n n^{-s}$. The prototypical Dirichlet series is the series $\sum n^{-s}$ representing the zeta function. Dirichlet series are important in number theory. They were used by Dirichlet to study the distribution of prime numbers in arithmetic progressions.

The modulus $|a_n n^{-(\sigma+it)}| = |a_n| n^{-\sigma}$ of the n th term of the series depends only on the real part σ of $s = \sigma + it$. The modulus decreases as σ increases. Thus if the series (4.1) converges absolutely for $s = s_0$, then it converges absolutely for all s satisfying $\operatorname{Re} s \geq \operatorname{Re} s_0$. By the Weierstrass M -test, it also converges uniformly in the half-plane $\{\operatorname{Re} s \geq \operatorname{Re} s_0\}$, by comparison with the numerical series $\sum |a_n| n^{-\operatorname{Re} s_0}$. Proceeding in analogy with the definition of radius of convergence of a power series, we define σ_a to be the infimum of σ for which $\sum |a_n| n^{-\sigma}$ converges, and we have the following.

Theorem. *For each Dirichlet series (4.1), there is a unique extended real number σ_a , $-\infty \leq \sigma_a \leq +\infty$, such that the series converges absolutely if $\operatorname{Re} s > \sigma_a$, and the series does not converge absolutely if $\operatorname{Re} s < \sigma_a$. For any $\sigma_0 > \sigma_a$, the series converges uniformly on the half-plane $\{\operatorname{Re} s \geq \sigma_0\}$. The sum $f(s) = \sum a_n n^{-s}$ is analytic for $\operatorname{Re} s > \sigma_a$.*

The extended real number σ_a is called the **abscissa of absolute convergence** of the series (4.1).

Example. The series $\sum n^{-s}$ for $\zeta(s)$ has abscissa of absolute convergence $\sigma_a = 1$. It does not converge at $s = 1$.

Suppose the function $f(s)$ is represented by a Dirichlet series, say $f(s) = \sum a_n n^{-s}$ for $\operatorname{Re} s > \sigma_a$. Since the series converges uniformly in some half-plane, and each of the terms $a_n n^{-s} \rightarrow 0$ as $\sigma \rightarrow \infty$ for $n \geq 2$, we have

$$\sum_{n=2}^{\infty} \frac{a_n}{n^s} \rightarrow 0 \quad \text{as} \quad \sigma = \operatorname{Re} s \rightarrow \infty.$$

Consequently, $f(s) \rightarrow a_1$ as $\sigma \rightarrow \infty$. Similarly, we can capture the other coefficients of the series from $f(s)$. We write

$$(4.2) \quad n^s \left[f(s) - a_1 - \frac{a_2}{2^s} - \cdots - \frac{a_{n-1}}{(n-1)^s} \right] = a_n + \sum_{k=n+1}^{\infty} a_k \frac{n^s}{k^s}.$$

To estimate the series on the right, choose σ_0 large so that $\sum |a_k| k^{-\sigma_0} = M < \infty$. Then

$$\left| \sum_{k=n+1}^{\infty} a_k \frac{n^s}{k^s} \right| \leq \sum_{k=n+1}^{\infty} \frac{n^{\sigma}}{k^{\sigma-\sigma_0}} \frac{|a_k|}{k^{\sigma_0}} \leq \frac{n^{\sigma}}{(n+1)^{\sigma-\sigma_0}} M.$$

Since $n^{\sigma}/(n+1)^{\sigma-\sigma_0} \rightarrow 0$ as $\sigma \rightarrow \infty$, the series on the right-hand side of (4.2) tends to 0 as $\sigma \rightarrow \infty$, and the left-hand side of (4.2) tends to a_n as $\sigma \rightarrow \infty$. Thus starting with a_1 , we can determine in succession the coefficients a_n from $f(s)$. In particular, the coefficients a_n are determined uniquely by the function $f(s)$, and we have proved the first statement of the following theorem.

Theorem. *If an analytic function $f(s)$ can be represented by a Dirichlet series in some half-plane $\{\operatorname{Re} s > \sigma_0\}$, then that Dirichlet series is unique. If $f(s) = \sum a_n n^{-s}$ and $g(s) = \sum b_n n^{-s}$ are represented by Dirichlet series in a half-plane, then so are their sum $f(s) + g(s)$ and their product $f(s)g(s)$. Further, the Dirichlet series of the product $f(s)g(s)$ is $\sum c_n n^{-s}$, where*

$$c_n = \sum_{d|n} a_d b_{n/d}.$$

Here the notation “ $d|n$ ” means “ d divides n .” The theorem follows by multiplying the series for $f(s)$ and $g(s)$ term by term and regrouping the terms, which we can do for absolutely convergent series.

The relation between convergence and absolute convergence of Dirichlet series is more delicate than the corresponding relationship for power series. It may occur that a Dirichlet series converges not only for $\sigma > \sigma_a$, but also for $\sigma > \sigma_a - 1$.

Example. The Dirichlet series $\sum (-1)^n n^{-s}$ has $\sigma_a = 1$. However, it converges whenever $s = \sigma$ is real and positive, by the alternating series test. The following theorem shows that it converges at every point of the right half-plane $\{\operatorname{Re} s > 0\}$.

Theorem. *If the Dirichlet series $\sum a_n n^{-s}$ converges at a point $s = s_0$, then it converges for all s such that $\operatorname{Re} s > \operatorname{Re} s_0$. Further, it converges uniformly in any sector $\{|\arg(s - s_0)| \leq (\pi/2) - \varepsilon\}$ with vertex s_0 and aperture strictly less than π .*

This is proved using Abel's formula for summation by parts. Let $\sum a_k$ be a series, with partial sums

$$A_n = \sum_{k=1}^n a_k.$$

The formula for summation by parts is analogous to the formula for integration by parts,

$$(4.3) \quad \sum_{k=m}^n a_k b_k = A_n b_n - A_{m-1} b_m + \sum_{k=m}^{n-1} A_k (b_k - b_{k+1}).$$

This can be checked by substituting $a_k = A_k - A_{k-1}$ and adjusting the indices.

Before applying the formula, we make two preliminary reductions in order to simplify the proof. First, observe that $\sum a_n n^{-s}$ converges at $s = s_1$ if and only if $\sum (a_n n^{-s_0}) n^{-s}$ converges at $s = s_1 - s_0$. Thus by replacing $\sum a_n n^{-s}$ by $\sum b_n n^{-s}$, where $b_n = a_n n^{-s_0}$, we can assume that $s_0 = 0$ and that $\sum a_n$ converges. The second reduction is to replace a_1 by $a_1 - \sum a_n$, so that we can assume that $\sum a_n = 0$. We must show that for any $C > 0$, the series $\sum a_n n^{-s}$ converges uniformly for $|s| \leq C\sigma$, where $s = \sigma + it$ and $\sigma > 0$. We do this by invoking the summation by parts formula, which in this case becomes

$$(4.4) \quad \sum_{k=m}^n \frac{a_k}{k^s} = \frac{A_n}{n^s} - \frac{A_{m-1}}{m^s} + \sum_{k=m}^{n-1} A_k \left(\frac{1}{k^s} - \frac{1}{(k+1)^s} \right).$$

Let ε_m be the maximum of $|A_k|$ for $k \geq m-1$. Then $\varepsilon_m \rightarrow 0$ as $m \rightarrow \infty$. From (4.4) we obtain

$$(4.5) \quad \left| \sum_{k=m}^n \frac{a_k}{k^s} \right| \leq \varepsilon_m \left(\frac{1}{n^\sigma} + \frac{1}{m^\sigma} + \sum_{k=m}^{n-1} \left| \frac{1}{k^s} - \frac{1}{(k+1)^s} \right| \right).$$

The first two summands are each bounded by 1, and

$$\begin{aligned} \left| \frac{1}{k^s} - \frac{1}{(k+1)^s} \right| &= |s| \left| \int_k^{k+1} r^{-s-1} dr \right| \leq |s| \int_k^{k+1} r^{-\sigma-1} dr \\ &= \frac{|s|}{\sigma} \left(\frac{1}{k^\sigma} - \frac{1}{(k+1)^\sigma} \right) = C \left(\frac{1}{k^\sigma} - \frac{1}{(k+1)^\sigma} \right). \end{aligned}$$

If we substitute these estimates in (4.5), we obtain a telescoping series,

$$\left| \sum_{k=m}^n \frac{a_k}{k^s} \right| \leq \varepsilon_m (2 + C) \sum_{k=m}^{n-1} \left(\frac{1}{k^\sigma} - \frac{1}{(k+1)^\sigma} \right) \leq \varepsilon_m (2 + C).$$

Since this tends to 0 as $m, n \rightarrow \infty$, we obtain uniform convergence.

The preceding theorem allows us to define the **abscissa of convergence** of a Dirichlet series $\sum a_n n^{-s}$ to be the extended real number σ_c , $-\infty \leq$

$\sigma_c \leq +\infty$, such that the series converges if $\operatorname{Re} s > \sigma_c$ and does not converge if $\operatorname{Re} s < \sigma_c$. Clearly, $\sigma_c \leq \sigma_a$. It is easy to see (Exercise 3) that $\sigma_a \leq \sigma_c + 1$, and further, this estimate is sharp.

Example. Consider the series representation from Exercise 3.4,

$$(4.6) \quad (1 - 2^{1-s}) \zeta(s) = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots, \quad \operatorname{Re} s > 1.$$

The abscissa of absolute convergence of the series on the right is $\sigma_a = 1$. By the alternating series test, the series converges for any positive real number $s > 0$. Consequently, the abscissa of convergence is $\sigma_c = 0$, and the series converges in the right half-plane to an analytic function. By the uniqueness principle, the identity (4.6) then holds in the entire right half-plane.

Exercises for XIV.4

1. Show using (4.5) that $\zeta(\sigma) > 0$ for $0 < \sigma < 1$. Conclude that the only zeros of the zeta function on the real line are the trivial zeros at the points $s = -2, -4, -6, \dots$.
2. Show that if the Dirichlet series $f(s) = \sum a_n n^{-s}$ converges in some half-plane, and if the a_n 's are not all zero, then there is a half-plane $\{\operatorname{Re} s > \sigma_0\}$ on which $f(s)$ has no zeros.
3. Show that if the terms $a_n n^{-s}$ are bounded for $s = s_0$, then the Dirichlet series $\sum a_n n^{-s}$ converges absolutely for $\operatorname{Re} s > 1 + \operatorname{Re} s_0$. Deduce that $\sigma_a \leq \sigma_c + 1$.
4. A coefficient sequence $\{a_n\}$ is **multiplicative** if $a_1 = 1$ and $a_{mn} = a_m a_n$ whenever m and n are relatively prime. It is **strongly multiplicative** if $a_1 = 1$ and $a_{mn} = a_m a_n$ for all m and n . Show that if $\{a_n\}$ is multiplicative, then

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = \prod_p (1 + a_p p^{-s} + a_{p^2} p^{-2s} + \cdots), \quad \operatorname{Re} s > \sigma_a,$$

while if $\{a_n\}$ is strongly multiplicative, then

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = \prod_p (1 - a_p p^{-s})^{-1}, \quad \operatorname{Re} s > \sigma_a.$$

5. The **Möbius μ -function** is defined on positive integers n by $\mu(1) = 1$, $\mu(n) = (-1)^r$ if n is a product of r distinct primes, and $\mu(n) = 0$ otherwise. (a) Show that μ is multiplicative but not strongly multiplicative. (b) Show that

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}, \quad \operatorname{Re} s > 1.$$

(c) Show directly, or from the uniqueness of Dirichlet series, that

$$\sum_{d|n} \mu(d) = 0, \quad n \geq 2.$$

(d) Show that if $f(s) = \sum a_n n^{-s}$ where $\{a_n\}$ is strongly multiplicative, then $1/f(s) = \sum a_n \mu(n) n^{-s}$.

6. Show that if $f(s)$ has a Dirichlet series $\sum a_n n^{-s}$ that converges in some half-plane, and if $a_1 \neq 0$, then $1/f(s)$ has a Dirichlet series that converges in some half-plane. *Hint.* Define the coefficients of the series for $1/f(s)$ by induction on n , and show that the resulting series converges. This procedure is essentially equivalent to showing that the sequences $a = \{a_n\}$ with $a_1 \neq 0$, form an abelian group, whose group operation is the **Dirichlet convolution** defined by $(a * b)_n = \sum_{d|n} a_d b_{n/d}$.
7. Let $\varphi(n)$ be **Euler's totient function**, defined to be the number of positive integers not exceeding n that are relatively prime to n . Show that

$$\frac{\zeta(s-1)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s}, \quad \operatorname{Re} s > 2.$$

Hint. Use the fact that $\sum_{d|n} \varphi(d) = n$.

8. If $\sigma_a(n)$ denotes the sum of the a th powers of the divisors of n , then

$$\zeta(s)\zeta(s-a) = \sum_{n=1}^{\infty} \frac{\sigma_a(n)}{n^s}, \quad \operatorname{Re} s > 1, \operatorname{Re} s > a + 1.$$

9. Let $d(n)$ denote the number of divisors of n . Establish the following formulae, and determine where they are valid:

$$\zeta(s)^2 = \sum_{n=1}^{\infty} \frac{d(n)}{n^s}, \quad \frac{\zeta(s)^3}{\zeta(2s)} = \sum_{n=1}^{\infty} \frac{d(n^2)}{n^s}, \quad \frac{\zeta(s)^4}{\zeta(2s)} = \sum_{n=1}^{\infty} \frac{d(n)^2}{n^s}$$

10. Show that

$$\frac{\zeta'(s)}{\zeta(s)} = - \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s}, \quad \operatorname{Re} s > 1,$$

where $\Lambda(n) = \log p$ if n is a power of a prime number p , and $\Lambda(n) = 0$ otherwise.

11. Define $a_n = 1$ and $b_n = \pm 1$ if $n \equiv \pm 1 \pmod{6}$; otherwise, set $a_n = b_n = 0$. Define two “Dirichlet L -series” by

$$L_1(s) = \sum \frac{a_n}{n^s} = 1 + \frac{1}{5^s} + \frac{1}{7^s} + \frac{1}{11^s} + \cdots,$$

$$L_{-1}(s) = \sum \frac{b_n}{n^s} = 1 - \frac{1}{5^s} + \frac{1}{7^s} - \frac{1}{11^s} + \cdots.$$

- (a) Show that $\{a_n\}$ and $\{b_n\}$ are strongly multiplicative. (See Exercise 4.)
 (b) Show that $L_1(s)$ has $\sigma_a = \sigma_c = 1$, and that $L_1(s) \rightarrow +\infty$ as $s \rightarrow 1+$.
 (c) Show that $L_{-1}(s)$ has $\sigma_a = 1$ and $\sigma_c = 0$.
 (d) Show that $\log L_1(s) - \log L_{-1}(s)$ can be expressed as

$$\sum_{p \equiv -1 \pmod{6}} \log(1 + p^{-s}) - \sum_{p \equiv -1 \pmod{6}} \log(1 - p^{-s}),$$

and deduce from (b) that there are infinitely many primes congruent to -1 modulo 6.

- (e) Show that $\log L_1(s) + \log L_{-1}(s)$ can be expressed as

$$-2 \sum_{p \equiv 1 \pmod{6}} \log(1 - p^{-s}) - \sum_{p \equiv -1 \pmod{6}} \log(1 - p^{-2s}),$$

and deduce that there are infinitely many primes congruent to 1 modulo 6.

Remark. Dirichlet showed using this technique that if $1 \leq k < m$, and if k is relatively prime to m , then the arithmetic progression $\{k, k+m, k+2m, \dots\}$ contains infinitely many primes. Further, for fixed m the primes are distributed asymptotically evenly among such arithmetic progressions.

12. Suppose that the partial sums of the Dirichlet series $\sum a_n n^{-s}$ are bounded at $s = s_0$. Show that the series converges for $\operatorname{Re} s > \operatorname{Re} s_0$.
 13. Suppose $a_n \geq 0$ for all n , and the abscissa of absolute convergence σ_a for the Dirichlet series $f(s) = \sum a_n n^{-s}$ is finite. Show that $f(s)$ cannot be extended analytically to any neighborhood of σ_a .
 14. Denote by σ_c and σ_a the abscissas of convergence and of absolute convergence, respectively, for the Dirichlet series $\sum a_n n^{-s}$. Show that

$$\sigma_a - 1 \leq \limsup_{n \rightarrow \infty} \frac{\log |a_n|}{\log n} \leq \sigma_c.$$

15. Denote by σ_c the abscissa of convergence of the Dirichlet series $\sum a_n n^{-s}$. (a) Show that if $\sum a_n$ converges, and $R_n = a_n + a_{n+1} +$

$\dots$, then

$$\sigma_c = \limsup_{n \rightarrow \infty} \frac{\log |R_n|}{\log n}.$$

(b) Show that if $\sum a_n$ does not converge, and $A_n = a_1 + a_2 + \dots + a_n$, then

$$\sigma_c = \limsup_{n \rightarrow \infty} \frac{\log |A_n|}{\log n}.$$

5. The Prime Number Theorem

We say that two functions $f(x)$ and $g(x)$ are **asymptotic** as $x \rightarrow \infty$, written $f \sim g$, if $f(x)/g(x) \rightarrow 1$ as $x \rightarrow \infty$. Let $\pi(x)$ be the number of prime numbers p such that $p \leq x$. Thus $\pi(2) = 1$, $\pi(6) = 3$, $\pi(10) = 4$, and so on. The prime number theorem gives the asymptotic behavior of $\pi(x)$ as $x \rightarrow \infty$.

Theorem (Prime Number Theorem). *The number $\pi(x)$ of prime numbers not exceeding x satisfies $\pi(x) \sim x/\log x$ as $x \rightarrow \infty$.*

This formula for the asymptotic distribution of primes was surmised by several outstanding mathematicians, including Gauss. Tables of factorizations of the first several hundred thousand positive integers were already published before 1700. Gauss (or, more accurately, his assistant) calculated all primes up to about 3,000,000, and Gauss used the results to aid his researches into number theory.

After much effort by many mathematicians, the prime number theorem was proved in 1896 independently by Hadamard and de la Vallée Poussin. Their proofs were very difficult. Subsequently, several new proofs of the prime number theorem have been given, and the various proofs have been gradually simplified. We will follow D. Zagier's exposition of a proof due to D.J. Newman (1972). As with most proofs of the prime number theorem, the proof is divided into two parts. The first is to show that the zeta function has no zeros on the line $\{\operatorname{Re} s = 1\}$. The second is to establish a "Tauberian theorem" relating the zeta function and the distribution of primes.

An important role in the proof is played by the function $\Phi(s)$ defined by the Dirichlet series

$$(5.1) \quad \Phi(s) = \sum_p \frac{\log p}{p^s}, \quad \operatorname{Re} s > 1,$$

where as usual we use p for prime numbers. The series converges absolutely, and $\Phi(s)$ is analytic for $\operatorname{Re} s > 1$. If we differentiate logarithmically the

product formula (3.1) for the zeta function, we obtain

$$(5.2) \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \frac{\log p}{p^s - 1} = \Phi(s) + \sum \frac{\log p}{p^s(p^s - 1)}.$$

The series on the right converges to an analytic function for $\operatorname{Re} s > \frac{1}{2}$. Since $\zeta'(s)/\zeta(s)$ is meromorphic on the entire complex plane, (5.2) shows that $\Phi(s)$ extends to be meromorphic for $\{\operatorname{Re} s > \frac{1}{2}\}$, with simple poles at the poles and zeros of the zeta function. Thus $\Phi(s)$ has a simple pole at $s = 1$ with residue 1, and $\Phi(s)$ has a simple pole at $s = s_0$ with residue $-m$ if $\zeta(s)$ has a zero at $s = s_0$ of order m .

Theorem. *The meromorphic function $\Phi(s)$ has no poles on the vertical line $\{\operatorname{Re} s = 1\}$ except at $s = 1$. The zeta function $\zeta(s)$ has no zeros on the line $\{\operatorname{Re} s = 1\}$.*

The proof is by magic. Fix $t > 0$, and suppose that $\zeta(s)$ has order q at $1 + it$ and order q' at $1 + 2it$, where $q, q' \geq 0$. Then it has orders q and q' at $1 - it$, and $1 - 2it$, respectively. Consider

$$0 \leq \left(p^{it/2} + p^{-it/2}\right)^4 = p^{2it} + 4p^{it} + 6 + 4p^{-it} + p^{-2it}.$$

Multiplying by $\varepsilon(\log p)/p^{1+\varepsilon}$ and summing over the primes, we obtain

$$0 \leq \varepsilon[\Phi(1 + \varepsilon + 2it) + 4\Phi(1 + \varepsilon + it) + 6\Phi(1 + \varepsilon) + 4\Phi(1 + \varepsilon - it) + \Phi(1 + \varepsilon - 2it)].$$

Since the residues of $\Phi(s)$ at the points $1, 1 \pm it, 1 \pm 2it$ are respectively $1, -q, -q'$, we have $\varepsilon\Phi(1 + \varepsilon) \rightarrow 1$, $\varepsilon\Phi(1 + \varepsilon \pm it) \rightarrow -q$, and $\varepsilon\Phi(1 + \varepsilon \pm 2it) \rightarrow -q'$ as $\varepsilon \rightarrow 0$. In the limit we then obtain

$$0 \leq -q' - 4q + 6 - 4q - q' \leq 6 - 8q - 2q'.$$

Since q and q' are nonnegative integers, this can occur only when $q = 0$. Thus $1 + it$ is not a zero of $\zeta(s)$ nor a pole of $\Phi(s)$. It follows that $\Phi(s)$ has no poles on the line $1 + i\mathbb{R}$ except $s = 1$, and $\zeta(s)$ has no zeros on the line.

None of the proofs of the prime number theorem treat the function $\pi(x)$ directly. They treat the function

$$(5.3) \quad \vartheta(x) = \sum_{p \leq x} \log p.$$

The function $\vartheta(x)$ is an increasing, piecewise continuous function of x . Since there are at most $\pi(x)$ summands in (5.3),

$$(5.4) \quad 0 \leq \vartheta(x) \leq \pi(x) \log x, \quad x \geq 1.$$

For fixed $\varepsilon > 0$, we have

$$\begin{aligned}\vartheta(x) &\geq \sum \{\log p : x^{1-\varepsilon} < p \leq x\} \geq (1-\varepsilon)(\log x)(\pi(x) - \pi(x^{1-\varepsilon})) \\ &\geq (1-\varepsilon)(\log x)(\pi(x) - x^{1-\varepsilon}).\end{aligned}$$

Combining this with (5.4), we obtain

$$\frac{\vartheta(x)}{x} \leq \pi(x) \frac{\log x}{x} \leq \frac{1}{1-\varepsilon} \frac{\vartheta(x)}{x} + \frac{\log x}{x^\varepsilon}.$$

Since $(\log x)/x^\varepsilon \rightarrow 0$ as $x \rightarrow \infty$, we see from this that $\vartheta(x)/x \rightarrow 1$ if and only if $\pi(x)(\log x)/x \rightarrow 1$. We have proved the following.

Lemma. *The prime number theorem holds if and only if $\vartheta(x) \sim x$.*

It is clear that $\vartheta(x) \leq x \log x$. We need the following stronger estimate for the growth of $\vartheta(x)$, which goes back to Chebyshev.

Lemma. $\vartheta(x) \leq Cx$ for some constant C and $x \geq 0$.

To see this, we consider the binomial coefficient

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} < 2^{2n}.$$

Each prime number p between n and $2n$ divides the binomial coefficient. Consequently, their product does, and their product is bounded by 2^{2n} , so

$$\sum \{\log p : n < p < 2n\} \leq 2n \log 2.$$

Applying this estimate to $n = 2^k$ and summing, we obtain

$$\begin{aligned}\vartheta(2^m) &= \sum_{k=1}^m \sum \{\log p : 2^{k-1} < p < 2^k\} \\ &\leq 2(1 + 2^2 + \cdots + 2^{m-1}) \log 2 < 2^{m+1} \log 2.\end{aligned}$$

Given x , we choose m such that $2^{m-1} < x \leq 2^m$, and then $\vartheta(x) \leq \vartheta(2^m) \leq 2^{m+1} \log 2 \leq (4 \log 2)x$.

Next we connect $\Phi(s)$ and $\vartheta(t)$ via the Laplace transform. This requires a logarithmic change of variable, to convert the multiplication on $(0, \infty)$ to addition on $(-\infty, \infty)$. We consider the function $t \mapsto \vartheta(e^t)$, which is a piecewise continuous function of t .

Lemma. *The Laplace transform $(\mathcal{L} \vartheta(e^t))(s)$ of the function $\vartheta(e^t)$ converges absolutely for $\operatorname{Re} s > 1$, and*

$$(\mathcal{L} \vartheta(e^t))(s) = \frac{\Phi(s)}{s}, \quad \operatorname{Re} s > 1.$$

Since $\vartheta(e^t) \leq Ce^t$, the Laplace transform of $\vartheta(e^t)$ converges absolutely for $\operatorname{Re} s > 1$. Let p_n denote the n th prime. Then $\vartheta(e^t)$ is constant for $\log p_n < t < \log p_{n+1}$, so

$$\int_{\log p_n}^{\log p_{n+1}} e^{-st} \vartheta(e^t) dt = \vartheta(p_n) \frac{e^{-st}}{-s} \Big|_{\log p_n}^{\log p_{n+1}} = \frac{1}{s} \vartheta(p_n) (p_n^{-s} - p_{n+1}^{-s}).$$

We sum over the primes and use $\vartheta(p_n) - \vartheta(p_{n-1}) = \log p_n$ to obtain

$$\begin{aligned} \int_0^\infty e^{-st} \vartheta(e^t) dt &= \frac{1}{s} \sum \vartheta(p_n) (p_n^{-s} - p_{n+1}^{-s}) = \frac{1}{s} \sum (\vartheta(p_n) - \vartheta(p_{n-1})) p_n^{-s} \\ &= \frac{1}{s} \sum \frac{\log p_n}{p_n^s} = \frac{1}{s} \Phi(s), \quad \operatorname{Re} s > 1. \end{aligned}$$

Instead of dealing with the Laplace transform of $\vartheta(e^t)$, it will be convenient to treat the Laplace transform of the function

$$(5.5) \quad h(t) = \vartheta(e^t) e^{-t} - 1.$$

The effect of multiplying by e^{-t} is to translate the variable s of the Laplace transform by 1, and the effect of subtracting 1 is to kill the pole of $\Phi(s)$ at $s = 1$.

Lemma. *The function $h(t)$ defined by (5.5) is a bounded piecewise continuous function on $[0, \infty)$, whose Laplace transform converges absolutely in the right half-plane and satisfies*

$$(5.6) \quad (\mathcal{L}h)(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}, \quad \operatorname{Re} s > 0.$$

The Laplace transform $(\mathcal{L}h)(s)$ extends analytically across the imaginary axis.

The link between the analyticity of $\Phi(s)$ and the asymptotic behavior of $h(t)$ is contained in the following “Tauberian theorem.”

Theorem. *Let $h(t)$, $t \geq 0$, be a bounded piecewise continuous function whose Laplace transform $(\mathcal{L}h)(z)$ extends analytically across the imaginary axis. Then*

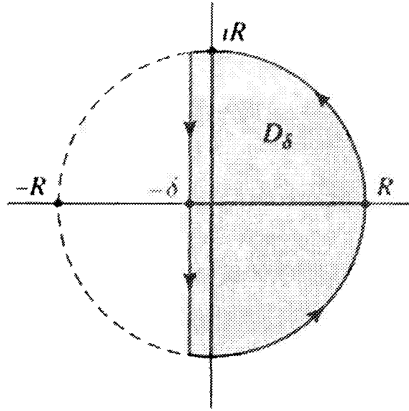
$$\lim_{T \rightarrow \infty} \int_0^T h(t) dt = \lim_{\varepsilon \rightarrow 0+} (\mathcal{L}h)(\varepsilon).$$

Note that the Laplace transform $(\mathcal{L}h)(z)$ is defined only for $\operatorname{Re} z > 0$, where the defining integral converges absolutely. Let $g(z)$ denote the analytic continuation of $(\mathcal{L}h)(z)$ across the imaginary axis. The theorem asserts that the Laplace transform exists, possibly as an improper integral, at $z = 0$, and is equal to $g(0)$.

For the proof, let $\varepsilon > 0$ be small. For fixed $T > 0$, define

$$g_T(z) = \int_0^T e^{-zt} h(t) dt, \quad z \in \mathbb{C}.$$

Then $g_T(z)$ is an entire function. We will show that $|g(0) - g_T(0)| < \varepsilon$ for large T .



Suppose $|h(t)| \leq M$ for $0 \leq t < \infty$. Choose $R > 0$ so large that $M/R < \varepsilon/4$, and then choose $\delta > 0$ very small, so that $g(z)$ is defined and analytic in and across the boundary of the domain D_δ consisting of z such that $|z| < R$ and $\operatorname{Re} z > -\delta$. From Cauchy's theorem we have

$$(5.7) \quad g(0) - g_T(0) = \frac{1}{2\pi i} \int_{\partial D_\delta} (g(z) - g_T(z)) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z}.$$

We break ∂D_δ into pieces. Let γ_+ be the semicircle $\{|z| = R, \operatorname{Re} z > 0\}$, let α_δ be the vertical interval along ∂D_δ where $\operatorname{Re} z = -\delta$, and let β_δ be the two short arcs of the circle $\{|z| = R\}$ connecting α_δ to $\pm iR$, all oriented according to the orientation of ∂D_δ . For $z = x + iy \in \gamma_+$ we have

$$|g(z) - g_T(z)| = \left| \int_T^\infty h(t) e^{-zt} dt \right| \leq M \int_T^\infty e^{-xt} dt = \frac{M e^{-xT}}{x}.$$

Since also $|1 + z^2/R^2| = 2|x|/R$ for $|z| = R$, we obtain from the ML -estimate that the right-hand side of (5.7), integrated over γ_+ , is bounded by

$$\begin{aligned} & \left| \frac{1}{2\pi i} \int_{\gamma_+} (g(z) - g_T(z)) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z} \right| \\ & \leq \frac{1}{2\pi} \frac{M e^{-xT}}{x} \cdot e^{xT} \cdot \frac{2x}{R} \cdot \frac{1}{R} \cdot \pi R = \frac{M}{R} < \frac{\varepsilon}{4}. \end{aligned}$$

For the integral of $g(z) - g_T(z)$ over the remainder of ∂D_δ , we treat the functions separately. Let γ_- be the semicircle $\{|z| = R, \operatorname{Re} z < 0\}$. Since

$g_T(z)$ is entire,

$$\frac{1}{2\pi i} \int_{\alpha_\delta \cup \beta_\delta} g_T(z) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z} = \frac{1}{2\pi i} \int_{\gamma_-} g_T(z) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z},$$

and we estimate the left-hand side just as for the first integral. If $z = x + iy \in \gamma_-$, then

$$|g_T(z)| = \left| \int_0^T h(t) e^{-zt} dt \right| \leq M \int_0^T e^{-xt} dt < \frac{M e^{-xT}}{|x|},$$

so

$$\left| \frac{1}{2\pi i} \int_{\gamma_-} g_T(z) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z} \right| \leq \frac{M}{R} < \frac{\varepsilon}{4}.$$

Now $|e^{zT}| \leq 1$ on β_δ and the length of β_δ tends to 0 with δ , so we can choose $\delta > 0$ so small that for any $T > 0$,

$$\left| \frac{1}{2\pi i} \int_{\beta_\delta} g(z) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z} \right| < \frac{\varepsilon}{4}.$$

Finally, $|e^{zT}| = e^{-\delta T}$ on α_δ , so for T large the integral over α_δ is also bounded by $\varepsilon/4$. Adding the four estimates, we obtain $|g(0) - g_T(0)| < \varepsilon$ for T large, as required.

If we apply the Tauberian theorem to the function $h(t)$ defined by (5.5), we conclude that

$$\lim_{T \rightarrow \infty} \int_0^T [\vartheta(e^t) e^{-t} - 1] dt$$

exists. After a change of variable $x = e^t$, we find that

$$(5.8) \quad \lim_{R \rightarrow \infty} \int_1^R \left(\frac{\vartheta(x)}{x} - 1 \right) \frac{dx}{x}$$

exists. We complete the proof of the prime number theorem by showing that this implies $\vartheta(x) \sim x$.

Lemma. *If $\vartheta(x)$ is an increasing function of x such that the limit (5.8) exists, then $\vartheta(x) \sim x$.*

Suppose that $\vartheta(x) > (1 + \varepsilon)x$ for some $\varepsilon > 0$ and for arbitrarily large values of x . For any such x we have

$$\begin{aligned} \int_x^{(1+\varepsilon)x} \left(\frac{\vartheta(t)}{t} - 1 \right) \frac{dt}{t} &\geq \int_x^{(1+\varepsilon)x} \left(\frac{\vartheta(x)}{t} - 1 \right) \frac{dt}{t} \\ &\geq \int_x^{(1+\varepsilon)x} \left[(1 + \varepsilon) \frac{x}{t} - 1 \right] \frac{dt}{t} \\ &= \int_1^{1+\varepsilon} \left[\frac{1 + \varepsilon}{r} - 1 \right] \frac{dr}{r} = c > 0. \end{aligned}$$

Thus

$$\int_1^R \left(\frac{\vartheta(t)}{t} - 1 \right) \frac{dt}{t}$$

increases by c over infinitely many disjoint intervals of the form $(x, (1 + \varepsilon)x)$, so the integral cannot have a limit as $R \rightarrow \infty$. A similar calculation shows that we cannot have $\vartheta(x) < (1 - \varepsilon)x$ for arbitrarily large values of x . Consequently, $\vartheta(x)/x \rightarrow 1$ as $x \rightarrow \infty$.

Exercises for XIV.5

1. Why is the binomial coefficient “ m choose k ” less than 2^m ?
2. Show that $|1 + z^2/R^2| = 2|\operatorname{Re} z|/R$ for $|z| = R$.
3. Compute $\pi(x)$ for $x = 10$ and $x = 100$, and compare the result with $x/\log x$.
4. Show that $\pi(x) \sim \int_2^x (1/\log t) dt$. *Remark.* This asymptotic formula (with equivalent notation) was found scribbled on the back page of a table of logarithms that Gauss was given at age fourteen.
5. Show that if p_n is the n th prime, then $p_n \sim n \log n$ as $n \rightarrow \infty$.
6. Show that $\sum 1/p = +\infty$, where the sum is over the primes, by filling in the details of the following elementary argument. If the sum converges, choose N such that the tail $S = \sum_{k>N} 1/p_k$ satisfies $S < 1$. Show that $S + S^2 + S^3 + \cdots$ dominates the sum of $1/n$ over positive integers not divisible by $p_1, p_2, \dots, p_N$; hence it dominates $\sum_{m=1}^{\infty} 1/(1 + mp_1 p_2 \cdots p_N)$, which diverges.
7. Exactly where in the proof of the prime number theorem is the Chebyshev estimate $\vartheta(x) \leq Cx$ absolutely required? Can you modify the proof so that it is not required?
8. Let $h(t)$, $0 \leq t < \infty$, be a piecewise continuous function such that

$$H(T) = \int_0^T e^{-z_0 t} h(t) dt, \quad T \geq 0,$$

is bounded. Show that $(\mathcal{L}h)(z) = \int_0^\infty e^{-zt} h(t) dt$ exists as an improper integral for all z satisfying $\operatorname{Re} z > \operatorname{Re} z_0$. *Hint.* Show that

$$\int_0^T e^{-zt} h(t) dt = e^{-(z-z_0)T} H(T) + (z - z_0) \int_0^T e^{-(z-z_0)t} H(t) dt.$$

The Dirichlet Problem

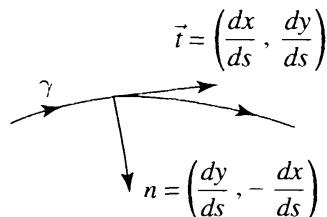
In Chapter X we used the Poisson kernel to solve the Dirichlet problem for the unit disk. In this chapter we study the Dirichlet problem for more general domains in the plane. The basic method, due to O. Perron, is to look for the solution of the Dirichlet problem as the upper envelope of a family of subsolutions. In Section 2 we introduce subharmonic functions, which play the role of the subsolutions. In Section 3 we derive Harnack's inequality, which provides a compactness criterion for families of harmonic functions. Perron's procedure for solving the Dirichlet problem is developed in Section 4. We apply the method to give another proof of the Riemann mapping theorem in Section 5. In Sections 6 and 7 we introduce Green's function.

1. Green's Formulae

In this section we will establish three fundamental formulae for dealing with harmonic and subharmonic functions. All are consequences of Green's theorem. We assume that the boundary curves and functions we treat are smooth enough so that Green's theorem is valid.

Let γ be a smooth curve in the plane. As before, we denote by $\mathbf{n}$ the unit normal vector to the curve. If γ is parameterized by arc length, $\gamma(s) = (x(s), y(s))$, then $\mathbf{n} = (dy/ds, -dx/ds)$. We denote the derivative of a smooth function $v(x, y)$ in the normal direction by $\partial v / \partial n$. Thus

$$\frac{\partial v}{\partial n} = \nabla v \cdot \mathbf{n} = \frac{\partial v}{\partial x} \frac{dy}{ds} - \frac{\partial v}{\partial y} \frac{dx}{ds}, \quad \frac{\partial v}{\partial n} ds = \frac{\partial v}{\partial x} dy - \frac{\partial v}{\partial y} dx.$$



Let D be a bounded domain with piecewise smooth boundary, and let u and v be smooth functions on $D \cup \partial D$. We consider integrals of the form

$$\int_{\partial D} u \frac{\partial v}{\partial n} ds = \int_{\partial D} -u \frac{\partial v}{\partial y} dx + u \frac{\partial v}{\partial x} dy.$$

By Green's theorem, this is equal to

$$\iint_D \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u \frac{\partial^2 v}{\partial x^2} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} + u \frac{\partial^2 v}{\partial y^2} \right) dx dy.$$

Rewriting this in terms of the gradient and Laplacian operators, we obtain **Green's first formula**,

$$(1.1) \quad \int_{\partial D} u \frac{\partial v}{\partial n} ds = \iint_D \nabla u \cdot \nabla v dx dy + \iint_D u \Delta v dx dy.$$

If we interchange u and v and subtract, the gradient terms cancel, and we obtain **Green's second formula**,

$$(1.2) \quad \int_{\partial D} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds = \iint_D (u \Delta v - v \Delta u) dx dy.$$

We will be particularly interested in this identity when one or both of u and v are harmonic. Then the corresponding Laplacian term is zero, and the formula simplifies. In the very special case that u is harmonic and $v \equiv 1$, the formula reduces to $\int_{\partial D} \partial u / \partial n ds = 0$, which was used in Chapter III.

Suppose $v(z)$ is harmonic in a punctured neighborhood of a point ζ . We say that $v(z)$ has a **logarithmic pole** at ζ if $v(z) - \log(1/|z - \zeta|)$ is harmonic at ζ . The following theorem provides a version of Green's second formula for functions with a logarithmic pole, which is analogous to Pompeiu's formula (Section IV.8) for the operator $\partial/\partial z$. It is sometimes referred to as **Green's third formula**.

Theorem. *Let D be a bounded domain with piecewise smooth boundary, and let u be a smooth function on $D \cup \partial D$. Let $\zeta \in D$, and let $v(z)$ be a harmonic function on $D \setminus \{\zeta\}$ such that v extends smoothly to ∂D and such that $v(z) - \log 1/|z - \zeta|$ is harmonic at ζ . Then*

$$(1.3) \quad u(\zeta) = -\frac{1}{2\pi} \iint_D v \Delta u dx dy - \frac{1}{2\pi} \int_{\partial D} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds.$$

Note that by Green's second formula, the right-hand side of (1.3) is zero if we replace $v(z)$ by a function that is harmonic on all of D . Thus we can add to $v(z)$ any harmonic function without affecting the right-hand side. We add the harmonic function $-v(z) + \log(1/|z - \zeta|)$ to $v(z)$, and we see that it suffices to establish (1.3) when $v(z)$ is replaced by $\log(1/|z - \zeta|)$.

So suppose $v(z) = \log(1/|z - \zeta|)$. We follow the idea of the proof of Pompeiu's theorem. (The picture is the same as the picture for the proof of

Cauchy's integral formula in Section IV.4.) We excise a small disk centered at ζ from D and set $D_\varepsilon = D \setminus \{|z - \zeta| \leq \varepsilon\}$, and we apply Green's second formula to D_ε . Taking into account $\Delta v = 0$ on D_ε , we obtain

$$\int_{\partial D} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds - \int_{|z|=\varepsilon} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds = - \iint_{D_\varepsilon} v \Delta u \, dx \, dy.$$

Here we have reversed the orientation of the circle $\{|z - \zeta| = \varepsilon\}$ to the usual counterclockwise orientation, so now for $z = \zeta + re^{i\theta}$ we have $\partial/\partial n = \partial/\partial r$. The logarithmic singularity at ζ is absolutely integrable,

$$\iint_{|z-\zeta|<1} \log \frac{1}{|z-\zeta|} \, dx \, dy = \int_0^{2\pi} \int_0^1 \left(\log \frac{1}{r} \right) r \, dr \, d\theta < \infty.$$

Consequently, the area integral over D_ε tends to the area integral over D as $\varepsilon \rightarrow 0$. To establish (1.3), it remains to show that

$$(1.4) \quad - \int_{|z-\zeta|=\varepsilon} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds \rightarrow 2\pi u(\zeta).$$

If $|(\nabla u)(z)| \leq M$ for z near ζ , then

$$\left| \int_{|z-\zeta|=\varepsilon} v \frac{\partial u}{\partial n} ds \right| \leq \left(\log \frac{1}{\varepsilon} \right) M \cdot 2\pi\varepsilon,$$

which tends to 0 with ε . Since $\frac{\partial v}{\partial n} = \frac{\partial}{\partial r} \log \frac{1}{r} = -\frac{1}{r}$ and $ds = r \, d\theta$, the other summand in (1.4) is

$$- \int_{|z-\zeta|=\varepsilon} u \frac{\partial v}{\partial n} ds = \int_0^{2\pi} u(\zeta + \varepsilon e^{i\theta}) d\theta.$$

Since the averages of $u(z)$ around small circles centered at ζ tend to $u(\zeta)$, we see that this tends to $2\pi u(\zeta)$ as $\varepsilon \rightarrow 0$. This establishes (1.4) and with it the formula (1.3).

Exercises for XV.1

1. Show that $v(z)$ has a logarithmic pole at ζ if and only if $v(z) = \log |f(z)|$ for some function $f(z)$ that is meromorphic near ζ and has a simple pole at ζ .
2. Let D be a bounded domain with smooth boundary, and let $u(z)$ be a smooth function on $D \cup \partial D$ such that $u(z) = 0$ for $z \in \partial D$. Show that $\iint_D u(z)(\Delta u)(z) \, dx \, dy \leq 0$, with strict inequality unless $u(z) = 0$ for all $z \in D$.
3. Let D be a bounded domain with smooth boundary, and let λ be a real number. Suppose there is a nonzero smooth function $u(z)$ on

$D \cup \partial D$ such that $\Delta u = \lambda u$ on D and $u = 0$ on ∂D . Show that $\lambda < 0$.

4. Let D be a bounded domain with smooth boundary, and let $u(z)$ and $v(z)$ be smooth functions on $D \cup \partial D$ such that $u(z)$ is harmonic on D and $v(z) = 0$ on ∂D . Show that $\iint_D \nabla u \cdot \nabla v \, dx \, dy = 0$.
5. Let D be a bounded domain with smooth boundary, and let $u(z)$ and $v(z)$ be smooth functions on $D \cup \partial D$ such that $u(z)$ is harmonic on D and $v(z) = u(z)$ on ∂D . Show that

$$\iint_D |\nabla v|^2 \, dx \, dy = \iint_D |\nabla u|^2 \, dx \, dy + \iint_D |\nabla(v - u)|^2 \, dx \, dy.$$

Remark. The quadratic form $\iint |\nabla v|^2 \, dx \, dy$ is called the **Dirichlet integral**. The identity shows that the solution to the Dirichlet problem minimizes the Dirichlet integral. This is an instance of **Dirichlet's principle**.

6. Let D be a bounded domain with smooth boundary, and let h be a continuous function on ∂D . The **Neumann problem** is to find a harmonic function u on D such that $\partial u / \partial n = h$ on ∂D . Show that the Neumann problem is not solvable unless $\int_{\partial D} h \, ds = 0$.
7. Let $h(e^{i\theta})$ be a smooth (say twice continuously differentiable) real-valued function on the unit circle, with Fourier series $h(e^{i\theta}) = \sum_{-\infty}^{\infty} b_n e^{in\theta}$. Show that the Neumann problem is solvable on the unit disk with boundary function $h(e^{i\theta})$ if and only if $b_0 = 0$. Express the solution explicitly as a Fourier series, and comment on the uniqueness of the solution.
8. Let $h(e^{i\theta})$ be a smooth real-valued function on the boundary of the annulus $\{\rho < |z| < \sigma\}$. Show that the Neumann problem is solvable with boundary function $h(z)$ if and only if $\int_0^{2\pi} h(\rho e^{i\theta}) \, d\theta = \int_0^{2\pi} h(\sigma e^{i\theta}) \, d\theta$. Express the solution explicitly as a Fourier series. (See Problem VI.1.6.)
9. Let $u(z)$ be a smooth function on the complex plane that is zero outside of some compact set. Show that

$$u(\zeta) = -\frac{1}{2\pi} \iint_{\mathbb{C}} (\Delta u)(z) \log \frac{1}{|z - \zeta|} \, dx \, dy, \quad \zeta \in \mathbb{C}.$$

Remark. If we integrate this formally by parts twice, we obtain

$$u(\zeta) = -\frac{1}{2\pi} \iint_{\mathbb{C}} u(z) \Delta \left(\log \frac{1}{|z - \zeta|} \right) \, dx \, dy, \quad \zeta \in \mathbb{C}.$$

Thus the “distribution” Laplacian of $-\frac{1}{2\pi} \log \frac{1}{|z - \zeta|}$ with respect to z is the point mass at ζ (the “Dirac delta-function”), in the sense that it is equal to 0 away from ζ , and it is infinite at ζ in such a way that its integral (total mass) is equal to 1.

2. Subharmonic Functions

Let D be a domain, and let $u(z)$ be a continuous function from D to the extended real line $[-\infty, \infty)$. We say that $u(z)$ is **subharmonic** if for each $z_0 \in D$, there is $\varepsilon > 0$ such that $u(z)$ satisfies the mean value inequality

$$(2.1) \quad u(z_0) \leq \int_0^{2\pi} u(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}, \quad 0 < r < \varepsilon.$$

If $u(z_0) = -\infty$, this inequality is automatically satisfied. Note also that ε may depend on z_0 . This definition of subharmonicity is the same as the definition of the mean value property (Section III.4), except that “=” is replaced by “ $\leq$ ” and the function $u(z)$ is allowed to assume the value $-\infty$.

Example. Any harmonic function is subharmonic. If $u(z)$ is harmonic, then equality holds in (2.1) for all r such that the closed disk $\{|z - z_0| \leq r\}$ is contained in D .

Example. If $f(z)$ is analytic, then $\log |f(z)|$ is subharmonic. More generally, if $u(z)$ is harmonic, or even just subharmonic, on the open set where $u(z) > -\infty$, then $u(z)$ is subharmonic.

Example. If $f(z)$ is analytic, then $|f(z)|$ is subharmonic. In this case the integral representation $f(z_0) = \int_0^{2\pi} f(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}$ and an obvious estimate yield (2.1) for $|f(z)|$.

Subharmonicity is analogous in many respects to convexity of functions of one real variable. A smooth function $\chi(t)$ is convex if and only if $\chi''(t) \geq 0$. (See Exercise 8.) There is a corresponding characterization of subharmonic functions in which the operator d^2/dt^2 is replaced by the Laplacian Δ .

Theorem. A smooth real-valued function $u(z)$ is subharmonic on D if and only if $\Delta u \geq 0$ on D .

This follows from Green’s third formula (1.3), applied to the functions $u(z)$ and $v(z) = \log(\rho/|z - z_0|)$ on a disk $\{|z - z_0| \leq \rho\}$, which we assume is contained in D . In this case, formula (1.3) becomes

$$u(z_0) = -\frac{1}{2\pi} \iint_{|z-z_0| \leq \rho} (\Delta u)(z) \log \left(\frac{\rho}{|z - z_0|} \right) dx dy + \int_0^{2\pi} u(z_0 + \rho e^{i\theta}) \frac{d\theta}{2\pi}$$

This shows that the mean value inequality holds for all $z_0 \in D$ and all sufficiently small $\rho > 0$ if and only if $\Delta u \geq 0$.

We list several simple consequences of the definition of subharmonicity.

Lemma. *If $u(z)$ and $v(z)$ are subharmonic, then*

- (i) $u(z) + v(z)$ is subharmonic,
- (ii) $c u(z)$ is subharmonic for any constant $c \geq 0$,
- (iii) $w(z) = \max(u(z), v(z))$ is subharmonic.

To check (iii), for instance, let $z_0 \in D$, and suppose the mean value estimate (2.1) holds for $u(z_0)$ and $v(z_0)$. Then since $u(z) \leq w(z)$ and $v(z) \leq w(z)$, both $u(z_0)$ and $v(z_0)$ are bounded by $\int_0^{2\pi} w(z_0 + re^{i\theta}) d\theta / 2\pi$. Hence so is their maximum $w(z_0)$.

Both the maximum principle and the strict maximum principle hold for subharmonic functions. The proofs are virtually the same as those for harmonic functions.

Theorem (Strict Maximum Principle). *Let $u(z)$ be a subharmonic function on a domain D . If $u(z)$ attains its maximum at some point of D , then $u(z)$ is constant.*

To prove the strict maximum principle, one shows first that the set where $u(z)$ attains its maximum M is open. The proof of this given in Section III.5 carries over directly. It requires only the mean value inequality (2.1) in place of the mean value equality. The rest of the proof is easy. Since $u(z)$ is continuous, the set $\{u(z) < M\}$ is also open. Since D is connected, the set $\{u(z) = M\}$ must then be either empty or all of D .

Theorem (Maximum Principle). *Let D be a bounded domain, and let $u(z)$ be a subharmonic function on D . If $\limsup u(z) \leq 0$ as $z \rightarrow \partial D$, then $u(z) \leq 0$ for all $z \in D$.*

This follows from the strict maximum principle. We can argue as follows. The “lim sup” condition implies that the subharmonic function $w(z) = \max(u(z), 0)$ tends to 0 as z tends to ∂D . If we define $w(z)$ to be 0 on ∂D , then $w(z)$ is continuous on the compact set $D \cup \partial D$, so it attains its maximum at some point z_0 . If $w(z_0) > 0$, then $z_0 \in D$. By the strict maximum principle, $w(z) \equiv w(z_0)$ is constant, contradicting $w(z) = 0$ on ∂D . It follows that the maximum of $w(z)$ is 0, and consequently, $u(z) \leq 0$ on D .

The following corollary to the maximum principle is obtained by applying the maximum principle to the function $u(z) - v(z)$. It justifies the nomenclature “subharmonic.”

Theorem. Let D be a bounded domain, and let $u(z)$ be a subharmonic function on D . Let $v(z)$ be a continuous finite real-valued function on $D \cup \partial D$ that is harmonic on D . If $\limsup_{z \rightarrow \zeta} u(z) \leq v(\zeta)$ for all $\zeta \in \partial D$, then $u(z) \leq v(z)$ for all $z \in D$.

We conclude with a technique for constructing subharmonic functions that will play a key role in Perron's procedure for solving the Dirichlet problem.

Lemma. Let $u(z)$ be subharmonic on a domain D . Let $D_0 = \{|z - z_0| < \rho\}$ be a disk such that $D_0 \cup \partial D_0$ is contained in D and such that $u(z) \neq -\infty$ on ∂D_0 . Define the function $v(z)$ on D to be equal to $u(z)$ on $D \setminus D_0$ and to be the harmonic extension of $u|_{\partial D_0}$ on D_0 . Then $v(z)$ is subharmonic on D , and $u(z) \leq v(z)$ on D .

The proof is a simple application of the maximum principle. The function $v(z)$ on D_0 is the Poisson integral of the restriction of u to ∂D_0 , as in Section X.1. It is harmonic on D_0 and attains the boundary values $u(z)$ continuously on ∂D_0 . By the maximum principle, $u(z) \leq v(z)$ on D_0 . Consequently, $v(z)$ is continuous and satisfies $u(z) \leq v(z)$ on D . Evidently, $v(z)$ satisfies the mean value inequality at any point of $D \setminus D_0$, because $u(z)$ does and $u(z) = v(z)$ there. On D_0 , $v(z)$ has the mean value property, since it is harmonic there. Thus $v(z)$ is subharmonic on D .

Exercises for XV.2

1. Show that $u(z)$ is harmonic if and only if both $u(z)$ and $-u(z)$ are subharmonic.
2. Suppose $u(z)$ is subharmonic on a domain D . Show that the mean value inequality (2.1) holds for all r such that the disk $\{|z - z_0| \leq r\}$ is contained in D .
3. Let $u(z)$ be a continuous function from D to $[-\infty, \infty)$. Suppose $u_n(z)$ is an increasing sequence of subharmonic functions on D such that $u_n(z) \rightarrow u(z)$ for $z \in D$. Show that $u(z)$ is subharmonic. *Hint.* The difficulty is that the ε in (2.1), in the definition of subharmonicity, is allowed to depend on n . To circumvent this difficulty, use the result of Exercise 2.
4. Let $u(z)$ be a continuous function from D to $[-\infty, \infty)$. Suppose $u_n(z)$ is a decreasing sequence of subharmonic functions on D such that $u_n(z) \rightarrow u(z)$ for $z \in D$. Show that $u(z)$ is subharmonic.
5. Let $u(x, y)$ be a real-valued function on D with continuous first- and second-order partial derivatives. (a) For each $z_0 \in D$, show